

Basic Algebraic Number Theory

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Introduction

0.1 Notation

Here is some notation that may differ from other texts and that is consistently used throughout these notes:

- If we write $A \subset B$, we allow $A = B$. If we want to write that A is a proper subset of B , we write $A \subsetneq B$.
- $R^\times$ denotes the set of invertible elements of a ring R .
- If L, K are fields, we use the symbol L/K (read " L over K ") to denote that L is a field extension over K .
- We use the symbol $\#$ to denote the number of elements in a set e.g. $\#G$ is the number of elements of G .

0.2 Motivation

To motivate our interest in some of the topics we will study, we will attempt to provide an outline of a proof of Fermat's Last Theorem (or FLT for short). Let us first remind ourselves what the statement of this theorem is.

Theorem (Fermat's Last Theorem). *For $n \geq 3$, the equation*

$$x^n + y^n = z^n$$

has no non-trivial (meaning $x, y, z \neq 0$) solutions in $\mathbb{Z}$.

"Proof":

Let $n \geq 3$ and assume towards contradiction that the triple (x, y, z) solves the equation. WLOG we can assume that x, y, z are coprime. We can rewrite the equation as:

$$y^n = z^n - x^n = (z - x)(z - \zeta_n x) \dots (z - \zeta_n^{n-1} x),$$

where ζ_n denotes the n -th root of unity $\zeta_n = e^{\frac{2\pi i}{n}}$. The rest of the proof is divided into the following steps.

Step A:

The terms $(z - x)(z - \zeta_n x) \dots (z - \zeta_n^{n-1} x)$ are (essentially) coprime in $\mathbb{Z}[\zeta_n]$.

Step B:

We factor the terms into irreducibles

$$\begin{aligned} z - x &= \pi_1^{a_1} \dots \pi_k^{a_k} \\ z - \zeta_n x &= \rho_1^{b_1} \dots \rho_l^{b_l} \\ &\vdots \end{aligned}$$

Step A implies that π_i 's and ρ_j 's are pairwise distinct. Now we can rewrite the LHS like this

$$y^n = \pi_1^{na'_1} \dots \rho_1^{nb'_1} \dots$$

But then all the exponents on the RHS are divisible by n . This allows us to move onto the next step.

Step C:

We can now write

$$\begin{aligned} z - x &= \alpha^n \\ z - \zeta_n x &= \beta^n \\ &\vdots \end{aligned}$$

for some $\alpha, \beta \in \mathbb{Z}[\zeta_n]$.

Step D:

Our aim now is to write $\alpha, \beta, \dots$ in some basis and compare the coefficients to arrive at a contradiction.

Now it is important to note that this proof would not be a viable proof of Fermat's Last Theorem in general case, but if we address the problems in this proof, we can make it work in a certain special case.

Problems:

- Firstly, it is useful to assume that n is an odd prime. Once the proof is done for primes, one easily proves the theorem for composite numbers. Let us denote this odd prime as p .
- Now we should distinguish so called 1st and 2nd case of FLT. The 1st case is when p does not divide xyz and the 2nd case is when it does.
- Since $\mathbb{Z}[\zeta_n]$ need not be a UFD in general, the factorization into irreducibles in Step B may not be unique. We should instead factor into prime ideals.
- After factorizing into prime ideals, we will only get $z - x = A^p$ etc. for some ideal A in Step C. We need to introduce a "class group" which will be trivial iff $\mathbb{Z}[\zeta_n]$ is UFD. If we then limit ourselves to primes that do not divide the size of this class group (we will call them regular primes), then we can replace the ideal A with an element (after considering invertible elements).

After solving these problems, we will be able to prove the 1st case of Fermat's Last Theorem for regular primes. This way the proof serves as a motivation for us to study the class group and other related topics.

0.3 Recap

In this section we state important definitions and theorems from the introductory Commutative Algebra course we will need. All proofs and further details can be found in the text linked below (in Czech only):

<https://karlin.mff.cuni.cz/~kala/files/UKA24.pdf>

We may also sometimes need to reference theorems or propositions, which can be found in this text. These will be marked by the number of the theorem preceded by "ÚKA" e.g. "ÚKA Thm. 1.1".

Definition. A *number field* K is a finite degree extension of $\mathbb{Q}$.

Theorem (Primitive element theorem). *Let K be a number field. Then there exists an algebraic element $\alpha \in K$ such that $K = \mathbb{Q}(\alpha) = \mathbb{Q}[\alpha]$.*

Definition. Let $K \supset \mathbb{Q}$ be a field. Then

- (a) Let $S \supset R$ be rings. We say that $\alpha \in S$ is *integral* over R if there exists a monic polynomial $f \in R[x]$ such that $f(\alpha) = 0$.
- (b) $\mathcal{O}_K$ denotes the ring of elements $\alpha \in K$ that are integral over $\mathbb{Z}$.

Theorem. Let $S \supset R$ be rings, then TFAE:

- (a) $\alpha \in S$ is integral over R .
- (b) Minimal polynomial for α over R is monic.
- (c) $R[\alpha]$ is finitely generated R -module.
- (d) There exists R' such that $R[\alpha] \subset R' \subset S$ and R' is a finitely generated R -module.

Definition. Let $D \in \mathbb{Z}$ be such that $D \neq 0, 1$ and is square-free. A *quadratic field* is then $K = \mathbb{Q}(\sqrt{D})$.

Theorem. Let $K = \mathbb{Q}(\sqrt{D})$ be a quadratic field, then

$$\mathcal{O}_K = \begin{cases} \mathbb{Z}[\sqrt{D}] & \text{for } D \equiv 2, 3 \pmod{4} \\ \mathbb{Z}[\frac{1+\sqrt{D}}{2}] & \text{for } D \equiv 1 \pmod{4} \end{cases}$$

Definition. Let $K = \mathbb{Q}(\sqrt{D})$ be a quadratic field. If we write the elements of K like this

$$K = \{a + b\sqrt{D} \mid a, b \in \mathbb{Q}\},$$

then we define the *norm* as the map

$$\begin{aligned} N : K &\rightarrow \mathbb{Q} \\ a + b\sqrt{D} &\mapsto a^2 - b^2D. \end{aligned}$$

and the *trace* as the map

$$\begin{aligned} \text{Tr} : K &\rightarrow \mathbb{Q} \\ a + b\sqrt{D} &\mapsto 2a. \end{aligned}$$

1. Prime Ideals in $\mathcal{O}_K$

1.1 Trace & Norm

Definition. Let $L \supset K$ be a field extension of finite degree n . Let $e = (e_1, \dots, e_n)$ denote the basis of the K -vector space L and let $\alpha \in L$, then

(a) we denote by $M_e(\alpha)$ the *matrix of multiplication by α* defined as

$$M_e(\alpha) = (a_{ij})_{1 \leq i, j \leq n}$$

where

$$\alpha \cdot e_j = \sum_{i=1}^n a_{ij} e_i.$$

(b) we define the *norm* denoted by $N_{L/K}$ as

$$N_{L/K}(\alpha) = \det(M_e(\alpha)) \in K.$$

(c) we define the *trace* denoted by $\text{Tr}_{L/K}$ as

$$\text{Tr}_{L/K}(\alpha) = \text{Tr}(M_e(\alpha)) \in K.$$

(d) we define the *characteristic polynomial* for α in L/K as

$$c_{\alpha, L/K}(x) := \det(xI - M_e(\alpha)) = \sum_{i=0}^n c_i x^i \in K[x].$$

Remark. Let $\alpha \in L$ and $c_{\alpha, L/K}(x) = \sum_{i=0}^n c_i x^i$, then

(a) $c_n = 1$

(b) $c_0 = c_{\alpha, L/K}(0) = (-1)^n N_{L/K}(\alpha)$

(c) $c_{n-1} = -\text{Tr}_{L/K}(\alpha)$

Exercise. Prove that the trace and norm are independent of basis for L .

Example. Let $L = \mathbb{Q}(\sqrt{D})$ for some square-free $D \neq 0, 1$ and $K = \mathbb{Q}$. The basis for L is then $e = (1, \sqrt{D})$. For some $\alpha = a + b\sqrt{D} \in L$ we compute

$$\begin{aligned} \alpha \cdot 1 &= a + b\sqrt{D} \\ \alpha \cdot \sqrt{D} &= bD + a\sqrt{D} \end{aligned}$$

and now

$$\begin{aligned} M_e(\alpha) &= \begin{pmatrix} a & bD \\ b & a \end{pmatrix} \\ N_{L/K}(\alpha) &= a^2 - b^2 D = \alpha \alpha' \\ \text{Tr}_{L/K}(\alpha) &= 2a = \alpha + \alpha' \\ c_{\alpha}(x) &= x^2 - x \text{Tr}(\alpha) + N(\alpha) \end{aligned}$$

where $\alpha' = a - b\sqrt{D}$. Also if $\alpha \notin \mathbb{Q}$, then c_{α} is the minimal polynomial for α .

Proposition 1.1. Let $L \supset K$ be a field extension of finite degree n and $\alpha \in L$. Let us denote by $m(x)$ the minimal polynomial for α over K and by $c(x)$ the characteristic polynomial for α . Then

- (a) $c(x) = m^d(x)$ where $d = [L : K(\alpha)]$
- (b) $c(\alpha) = 0$
- (c) $N_{L/K}(\alpha) = (-1)^n c_0 = (-1)^n m_0^d = (N_{K(\alpha)/K}(\alpha))^d$
- (d) $\text{Tr}_{L/K}(\alpha) = -c_{n-1} = d \text{Tr}_{K(\alpha)/K}(\alpha)$

Proof. Parts (b)-(d) all follow from (a), so it suffices to prove part (a).

CASE 1: Assume $L = K(\alpha)$ for some $\alpha \in L$, then $d = 1$. We choose the basis $e = (1, \alpha, \dots, \alpha^{n-1})$ and compute the elements of $M_e(\alpha)$. We have

$$\begin{aligned} \alpha e_i &= \alpha^i = e_{i+1} && \text{for } i < n \\ \alpha e_n &= \alpha^n = -m_{n-1}\alpha^{n-1} - \dots - m_1\alpha - m_0 \end{aligned}$$

Now we can compute $c(x)$ by taking the determinant of the matrix

$$B := xI - M_e(\alpha) = \begin{pmatrix} x & 0 & 0 & \dots & m_0 \\ -1 & x & 0 & \dots & m_1 \\ 0 & -1 & x & \ddots & m_2 \\ \vdots & \ddots & \ddots & \ddots & \vdots \\ 0 & \dots & 0 & -1 & x + m_{n-1} \end{pmatrix}$$

which yields $c(x) = m(x)$ (left as an exercise).

CASE 2: Now let $d > 1$ and $(e_1, \dots, e_d)$ be a basis for $L/K(\alpha)$ and $(1, \alpha, \dots, \alpha^{\frac{n}{d}-1})$ be a basis for $K(\alpha)/K$. As an exercise, show that $e_j \alpha^i$ is a basis for L/K . Then

$$\alpha e_j \alpha^i = e_j \alpha^{i+1}$$

$$xI - M(\alpha) = \begin{pmatrix} B & & & \\ & B & & \\ & & \ddots & \\ & & & B \end{pmatrix}_{d \times d}$$

Taking the determinant yields

$$c(x) = (\det(B))^d = m^d(x).$$

□

Recall. Let K be a field and α be an algebraic and separable element over K and $m(x) := m_\alpha(x)$. Then a K -homomorphism

$$\varphi : K(\alpha) \rightarrow \bar{K}$$

is uniquely determined by $\varphi(\alpha)$. Moreover $\varphi(\alpha)$ is an (algebraic) conjugate of α i.e. it is a root of $m(x)$.

The separability of α is equivalent to m having no multiple roots, which happens iff the number of maps φ is equal to #of distinct roots of $m = \deg m = [K(\alpha) : K]$ iff

$$m(x) = \prod_{\varphi} (x - \varphi(\alpha))$$

Proposition 1.2. *Let L/K be finite separable extension and $\alpha \in L$, $c := c_{\alpha, L/K}$. Then*

(a) $c(x) = \prod (x - \varphi(\alpha))$

(b) $N_{L/K}(\alpha) = \prod \varphi(\alpha)$

(c) $\text{Tr}_{L/K}(\alpha) = \sum \varphi(\alpha)$

where the products and sums above iterate over all K -homomorphisms

$$\varphi : L \rightarrow \bar{L} = \bar{K}.$$

Proof. (a) The restriction $\varphi|_{K(\alpha)}$ is uniquely determined by $\varphi(\alpha)$, which is a conjugate of α . Conversely for a conjugate β there exists exactly one K -homomorphism

$$\begin{aligned} \psi : K(\alpha) &\rightarrow \bar{L} \\ \alpha &\mapsto \beta \end{aligned}$$

By ÚKA Cor. 2.10 ψ has exactly $d = [L : K(\alpha)]$ extensions φ . By Prop. 1.1

$$\begin{aligned} c(x) &= m^d(x) \\ &= \left(\prod_{\psi: K(\alpha) \rightarrow \bar{K}} (x - \psi(\alpha)) \right)^d \\ &= \prod_{\varphi: K(\alpha) \rightarrow \bar{K}} (x - \varphi(\alpha)). \end{aligned}$$

The rest follows from part (a) and Prop. 1.1. □

Exercise. Show, that $N_{L/K}(\alpha) \in K$ for all $\alpha \in L$.

Corollary 1.3. (a) *The norm as a map $N : L^\times \rightarrow K^\times$ is a homomorphism, therefore*

$$N(\alpha\beta) = N(\alpha)N(\beta) \quad \text{for all } \alpha, \beta \in L^\times$$

(b) *The trace as a map $\text{Tr} : L(+) \rightarrow K(+)$ is a group homomorphism, therefore*

$$\text{Tr}(\alpha + \beta) = \text{Tr}(\alpha) + \text{Tr}(\beta) \quad \text{for all } \alpha, \beta \in L$$

Exercise. Let $M \supset L \supset K$ be separable extensions of finite degree. Show, that then

(a) $N_{M/K} = N_{L/K} \circ N_{M/L}$

(b) $\text{Tr}_{M/K} = \text{Tr}_{L/K} \circ \text{Tr}_{M/L}$

(c) For $\alpha \in K$

$$\begin{aligned} N_{L/K}(\alpha) &= \alpha^n \\ \text{Tr}_{L/K}(\alpha) &= n\alpha, \quad n = [L : K] \end{aligned}$$

1.2 Discriminant

Assuming a separable extension L/K of finite degree and a tuple of elements from L , we would like to know, if said tuple is a suitable basis for the K -vector space L . The following notion of discriminant will help us in doing so.

Definition. Let L/K be a separable extension of degree n and $\alpha_1, \dots, \alpha_n \in L$, then we define the *discriminant* as

$$\Delta(\alpha_1, \dots, \alpha_n) := \det(\text{Tr}_{L/K}(\alpha_i \alpha_j))_{1 \leq i, j \leq n} \in K.$$

Lemma 1.4. *Let L/K be a separable extension of degree n and let $\varphi_1, \dots, \varphi_n$ be all the K -homomorphisms $L \rightarrow \bar{L}$ and $\alpha_1, \dots, \alpha_n \in L$. If we denote $M = (\varphi_i(\alpha_j))_{1 \leq i, j \leq n}$, then*

$$M^T M = \text{Tr}_{L/K}(\alpha_i \alpha_j)_{1 \leq i, j \leq n}.$$

and

$$\Delta(\alpha_1, \dots, \alpha_n) = (\det(M))^2.$$

Proof is left as an exercise (use Prop. 1.2).

Proposition 1.5. *Let $L = K(\gamma)$ be an extension of degree n over K . Then*

$$\Delta(1, \gamma, \dots, \gamma^{n-1}) = \left(\prod_{i < j} (\varphi_i(\gamma) - \varphi_j(\gamma)) \right)^2 = (-1)^{\binom{n}{2}} N_{L/K}(m'(\gamma)),$$

where m is the minimal polynomial for γ and φ_i 's are all the K -homomorphisms $L \rightarrow \bar{L}$.

Proof. We set

$$M := \begin{pmatrix} 1 = \varphi_1(1) & \varphi_1(\gamma) & \dots & \varphi_1(\gamma)^{n-1} \\ \vdots & \vdots & & \vdots \\ 1 = \varphi_n(1) & \varphi_n(\gamma) & \dots & \varphi_n(\gamma)^{n-1} \end{pmatrix}$$

Taking the determinant yields

$$\det(M) = \prod_{i < j} (\varphi_i(\gamma) - \varphi_j(\gamma)).$$

The first equality then follows from Lemma 1.4.

Let us now show the second equality.

$$\begin{aligned} m(x) &= \prod (x - \varphi_i(\gamma)) \\ m'(x) &= (x - \varphi_2(\gamma))(x - \varphi_3(\gamma)) \dots (x - \varphi_n(\gamma)) + \\ &\quad + (x - \varphi_1(\gamma))(x - \varphi_3(\gamma)) \dots (x - \varphi_n(\gamma)) + \dots \\ m'(\varphi_i(\gamma)) &= \prod_{j, j \neq i} (\varphi_i(\gamma) - \varphi_j(\gamma)). \end{aligned}$$

Now by Prop. 1.2

$$\begin{aligned} N_{L/K}(m'(\gamma)) &= \prod_i \varphi_i(m'(\gamma)) = \prod_i m'(\varphi_i(\gamma)) = \\ &= \prod_{i, j, i \neq j} (\varphi_i(\gamma) - \varphi_j(\gamma)) = \\ &= (-1)^{\binom{n}{2}} \prod_{i < j} (\varphi_i(\gamma) - \varphi_j(\gamma))^2. \end{aligned}$$

□

Observation 1. Since φ 's are distinct homomorphisms, then $\varphi_i(\gamma) \neq \varphi_j(\gamma)$ for all $i \neq j$ and hence

$$\Delta(1, \dots, \gamma^{n-1}) \neq 0.$$

Corollary 1.6. *Let L/K be a separable extension of degree n .*

(a) *The map*

$$\begin{aligned} B : L \times L &\rightarrow K \\ (\alpha, \beta) &\mapsto \text{Tr}(\alpha\beta) \end{aligned}$$

is a symmetric bilinear form, which is also non-degenerate i.e. for all $x \neq 0$ there exists y such that $B(x, y) \neq 0$.

(b) *Let $\alpha_1, \dots, \alpha_n, \beta_1, \dots, \beta_n \in L$, $\beta_j = \sum p_{ij}\alpha_i$, $P = (p_{ij}) \in K^{n \times n}$. Then*

$$\Delta(\beta_1, \dots, \beta_n) = (\det P)^2 \Delta(\alpha_1, \dots, \alpha_n)$$

(c) *$(\alpha_1, \dots, \alpha_n)$ is a basis for L over K iff $\Delta(\alpha_1, \dots, \alpha_n) \neq 0$.*

Proof. (a) Since separable extensions are simple, we find $\gamma \in L$ such that $K(\gamma) = L$ and then we can take the basis $(1, \gamma, \dots, \gamma^{n-1})$ for L/K . Then

$$\Delta(1, \gamma, \dots, \gamma^{n-1}) = \det(\text{Tr}(\gamma^i \gamma^j)),$$

which is not zero by Observation 1. The rest follows from the fact that bilinear form B is non-degenerate iff $\det(B(e_i, e_j)) \neq 0$ where e_i 's form a basis (exercise).

(b) If H is a matrix of bilinear form in basis $\alpha_1, \dots, \alpha_n$, then $P^T H P$ is the matrix in the basis $\beta_1, \dots, \beta_n$. Then

$$\begin{aligned} \Delta(\alpha_1, \dots, \alpha_n) &= \det(H) \\ \Delta(\beta_1, \dots, \beta_n) &= \det(P^T H P) = (\det P)^2 \det H. \end{aligned}$$

(c) Take $\beta_i := \gamma^{i-1}$ is a basis for $K(\gamma)/K$ and $\Delta(1, \gamma, \dots, \gamma^{n-1}) \neq 0$. Conversely if $(\alpha_1, \dots, \alpha_n)$ is not a basis, then the matrix P is singular, hence by part (b) we have $\Delta(\alpha_1, \dots, \alpha_n) = 0$. $\square$

Exercise. For $K := \mathbb{F}_2(x)$ and $L := K(\sqrt{x})$ consider the extension L/K . Show that then

$$\text{Tr}_{L/K}(\alpha) = 0 \quad \text{for all } \alpha \in L.$$

1.3 Integral basis

From this section onward, let K be a number field and we shall denote $n := [K : \mathbb{Q}]$. We will also consider ideals $I < \mathcal{O}_K$ to be non-zero. However we allow $I = \mathcal{O}_K$.

We are quite familiar with the basis for K over $\mathbb{Q}$ and how to find such basis. In this section, we will explore the notion of basis for $\mathcal{O}_K$ over $\mathbb{Z}$.

Observation 2. (a) $\forall \alpha \in K \exists m \in \mathbb{Z}, m \neq 0 : m\alpha \in \mathcal{O}_K$.

(b) For an ideal $I < \mathcal{O}_K$, we have $I \cap \mathbb{Z} \neq \{0\}$.

(c) If $\alpha \in \mathcal{O}_K$, then $N(\alpha), \text{Tr}(\alpha) \in \mathbb{Z}$.

(d) If $\alpha_1, \dots, \alpha_n \in \mathcal{O}_K$, then $\Delta(\alpha_1, \dots, \alpha_n) \in \mathbb{Z}$.

Proof. (a) Let

$$a_n \alpha^n + a_{n-1} \alpha^{n-1} + \dots + a_0 = 0 \quad a_i \in \mathbb{Z}, a_n \neq 0.$$

Then multiplying by a_n^{n-1} yields

$$(a_n \alpha)^n + a_{n-1} (a_n \alpha)^{n-1} + \dots + a_n^{n-1} a_0 = 0.$$

Hence $a_n \alpha \in \mathcal{O}_K$.

(b) Choose $0 \neq \alpha \in I \subset \mathcal{O}_K$ and write the minimal polynomial for α

$$\begin{aligned} 0 &= \alpha^n + a_{n-1}\alpha^{n-1} + \cdots + a_0 \quad a_i \in \mathbb{Z}, a_0 \neq 0 \\ a_0 &= -\alpha^n - a_{n-1}\alpha^{n-1} - \cdots - a_1\alpha \in I. \end{aligned}$$

(c) We will show the proof for norm, for trace the proof is analogous. $N(\alpha) \in \mathbb{Q}$. The conjugates $\varphi_i(\alpha)$ all lie in $\mathcal{O}_{\bar{\mathbb{Q}}}$ because they share the minimal polynomial with α . By Prop. 1.2 $N(\alpha) = \prod \varphi_i(\alpha)$, therefore $N(\alpha) \in \mathbb{Q} \cap \mathcal{O}_{\bar{\mathbb{Q}}} = \mathbb{Z}$.

(d) Trivially true. $\square$

Lemma 1.7. *Every ideal $I < \mathcal{O}_K$ contains some basis for $K/\mathbb{Q}$.*

Proof. Let $(\alpha_1, \dots, \alpha_n)$ be a basis for $K/\mathbb{Q}$. By part (a) of Observation 2 we can find non-zero $m \in \mathbb{Z}$ such that $m\alpha_i \in \mathcal{O}_K$ for all $1 \leq i \leq n$. By part (b) of Observation 2 we can choose non-zero $a \in I \cap \mathbb{Z}$. Then $(am\alpha_i)_{1 \leq i \leq n}$ is still a basis for $K/\mathbb{Q}$ and $am\alpha_i \in I$, which concludes the proof. $\square$

So far we know that the discriminant of a basis for $\mathcal{O}_K$ is non-zero and that it lies in $\mathbb{Z}$. Proposition 1.5 might suggest that it is also always positive, but the following example contradicts this conjecture.

Example. Let $K = \mathbb{Q}(i)$. We then have a basis $(1, i) =: (\alpha_1, \alpha_2)$. Now we compute its discriminant with the use of Prop. 1.5 for φ_1 being the identity and φ_2 the complex conjugation map.

$$\begin{aligned} \Delta(\alpha_1, \alpha_2) &= (i - \bar{i})^2 \\ \Delta(\alpha_1, \alpha_2) &= -4 \end{aligned}$$

Definition. Let $(\alpha_1, \dots, \alpha_n)$ be a basis for $K/\mathbb{Q}$. We call it *the integral basis for $I < \mathcal{O}_K$* if

$$I = \mathbb{Z}\alpha_1 + \cdots + \mathbb{Z}\alpha_n.$$

The value of $\Delta(\alpha_1, \dots, \alpha_n)$ does not depend on the choice of basis and is called the *discriminant of the ideal I* and denoted by $\Delta(I)$. The number $\Delta(\mathcal{O}_K)$ is called the *discriminant of the field K* and is often denoted by Δ_K .

Proposition 1.8. *Let $I < \mathcal{O}_K$ and $(\alpha_1, \dots, \alpha_n) \in I$ be a basis for $K/\mathbb{Q}$ such that $|\Delta(\alpha_1, \dots, \alpha_n)|$ is minimal. Then it is an integral basis.*

Proof. $|\Delta(\alpha_1, \dots, \alpha_n)| \in \mathbb{N}$, so considering the minimal value makes sense. Let

$$\alpha = c_1\alpha_1 + \cdots + c_n\alpha_n \in I, \quad \text{where } c_i \in \mathbb{Q}.$$

Assume towards contradiction that $c_i \notin \mathbb{Z}$ for some i . WLOG $c_1 \notin \mathbb{Z}$.

$$c_1 = \lfloor c_1 \rfloor + \{c_1\} \quad \lfloor c_1 \rfloor \in \mathbb{Z}.$$

Consider the basis

$$\beta_1 := \alpha - \lfloor c_1 \rfloor \alpha_1, \beta_2 := \alpha_2, \dots, \beta_n := \alpha_n.$$

We have

$$\beta_1 = \{c_1\}\alpha_1 + \cdots + c_n\alpha_n = \alpha - \lfloor c_1 \rfloor \alpha_1$$

and the transition matrix

$$\begin{pmatrix} \{c_1\} & c_2 & c_3 & \cdots & c_n \\ 0 & 1 & 0 & \cdots & 0 \\ \vdots & \ddots & \ddots & & \\ & & & & 1 \end{pmatrix}$$

Taking the determinat yields $\det = \{c_1\} \neq 0$. We have

$$|\Delta(\beta_1, \dots, \beta_n)| = \{c_1\}^2 |\Delta(\alpha_1, \dots, \alpha_n)| < |\Delta(\alpha_1, \dots, \alpha_n)|,$$

which contradicts minimality, hence all $c_i \in \mathbb{Z}$ and

$$I \subset \mathbb{Z}\alpha_1 + \dots + \mathbb{Z}\alpha_n.$$

The converse inclusion follows trivially. □

Exercise. Prove the Stickelberger theorem:

For all number fields K ,

$$\Delta_K \equiv 0 \text{ or } 1 \pmod{4}.$$

Proposition 1.9. $\mathcal{O}_K/I$ is finite for all $I < \mathcal{O}_K$.

Proof. Choose non-zero $a \in I \cap \mathbb{Z}$. Since $(a) < I$, we have a surjection

$$\begin{aligned} \mathcal{O}_K/(a) &\twoheadrightarrow \mathcal{O}_K/I \\ \alpha + (a) &\mapsto \alpha + I. \end{aligned}$$

Let $\omega_1, \dots, \omega_n$ be an integral basis for $\mathcal{O}_K$, then

$$\left\{ \sum_{i=1}^n c_i \omega_i : 0 \leq c_i \leq a \right\}$$

is a set of representatives of the equivalence on $\mathcal{O}_K$ induced by the ideal (a) . Hence $\mathcal{O}_K/(a)$ is finite with a^n elements, hence $\mathcal{O}_K/I$ is also finite. □

Definition. Let R be a domain and F its field of fractions. We say that R is integrally closed over $\mathbb{Z}$ if whenever $\alpha \in F$ is integral over $\mathbb{Z}$, then $\alpha \in R$.

We say that a domain R is Dedekind if

- (a) R is noetherian,
- (b) Each prime ideal $P < R$ is maximal,
- (c) R is integrally closed.

Corollary 1.10. $\mathcal{O}_K$ is a Dedekind domain.

Proof. $\mathcal{O}_K$ being integrally closed over $\mathbb{Z}$ is trivially true from definition. It suffices to show that $\mathcal{O}_K$ is noetherian and that each prime ideal $P < \mathcal{O}_K$ is maximal. By Prop. 1.8, every idea $I < \mathcal{O}_K$ is finitely generated, therefore $\mathcal{O}_K$ is noetherian. Let $P < \mathcal{O}_K$ be a prime ideal, then $\mathcal{O}_K/P$ is a domain, which is finite by Prop. 1.9. Since finite domain is a field (exercise), then P is maximal. □

1.4 Ideal cancellation

Notation. For a set A and an element β we denote

$$\beta A = \{\beta a \mid a \in A\}.$$

If A is an ideal, then $\beta A = (\beta)A$, where (β) denotes the principal ideal generated by β .

Lemma 1.11. (a) Let $A < \mathcal{O}_K$, $\beta \in K$ such that $\beta A \subset A$. Then $\beta \in \mathcal{O}_K$.

(b) Let $A, B < \mathcal{O}_K$ such that $A = AB$. Then $B = \mathcal{O}_K$.

Proof. Take $(\alpha_1, \dots, \alpha_n)$ a basis for A .

(a) Consider the following system of linear equations for which α_i are non-zero solutions.

$$\beta\alpha_i = \sum a_{ij}\alpha_j \quad a_{ij} \in \mathbb{Z}.$$

By linear algebra $\det M = 0$, where

$$M = \begin{pmatrix} \beta - a_{11} & -a_{12} & \dots & -a_{1n} \\ -a_{21} & \beta - a_{22} & \dots & -a_{2n} \\ \vdots & & \ddots & \vdots \\ & & & \beta - a_{nn} \end{pmatrix}$$

The determinant of M is also a monic polynomial in β with $\mathbb{Z}$ -coefficients $0 = \det M = \beta^n + \dots$ and root in β , hence $\beta \in \mathcal{O}_K$.

(b) Analogously we consider the following system of linear equations

$$\alpha_i = \sum b_{ij}\alpha_j \quad b_{ij} \in B.$$

with the matrix

$$\begin{pmatrix} 1 - b_{11} & -b_{12} & \dots & -b_{1n} \\ -b_{21} & 1 - b_{22} & \dots & -b_{2n} \\ \vdots & & \ddots & \vdots \\ & & & 1 - a_{nn} \end{pmatrix}$$

whose determinant is again 0. We can also write

$$\det = 0 = 1 + \heartsuit$$

where each term in $\heartsuit$ contains some b_{ij} , hence $\heartsuit \in B$ and hence also $1 \in B \implies B = \mathcal{O}_K$. $\square$

Exercise. Lemma 1.11 does not hold in general. As an exercise, find examples of rings, for which it does not hold.

Proposition 1.12. *Let $A, B < \mathcal{O}_K$, $\omega \in \mathcal{O}_K$ such that $(\omega)A = BA$. Then $(\omega) = B$.*

Proof. Take $\beta \in B$. We have

$$(\omega)A \supset (\beta)A \implies \frac{\beta}{\omega}A \subset A.$$

Since $\frac{\beta}{\omega}$ lies in K , it then follows by Lemma 1.11 part (a) that $\frac{\beta}{\omega} \in \mathcal{O}_K$. Hence

$$\omega^{-1}B = \left\{ \frac{\beta}{\omega} \mid \beta \in B \right\} \subset \mathcal{O}_K$$

is an ideal. We now have $A = (\omega^{-1}B)A$. By Lemma 1.11 part (b) $\omega^{-1}B = \mathcal{O}_K$ and hence $(\omega) = B$. $\square$

1.5 Ideal classes

Definition. Let $A, B < \mathcal{O}_K$. We say that A and B are equivalent and write $A \sim B$ if there exist $\alpha, \beta \in \mathcal{O}_K \setminus \{0\}$ such that $(\alpha)A = (\beta)B$. Classes of this equivalence are called *ideal classes*. The *class number* of K is defined as the number of ideal classes and denoted by h_K .

Remark. $h_K = 1 \iff \mathcal{O}_K$ is PID $\iff \mathcal{O}_K$ is UFD.

Proposition 1.13. $\forall K \exists M \in \mathbb{Z}^+$ such that $\forall \alpha, \beta \in \mathcal{O}_K, \beta \neq 0 \exists t \in \mathbb{Z}, 1 \leq t \leq M \exists \omega, \delta \in \mathcal{O}_K$ such that

$$t\alpha = \omega\beta + \delta, \quad |N\delta| < |N\beta|.$$

Proof. Let $\gamma = \frac{\alpha}{\beta} \in K$. It suffices to show that $\exists t, \omega$ such that

$$|N(t\gamma - \omega)| < 1.$$

Let $(\omega_1, \dots, \omega_n)$ be an integral basis for $\mathcal{O}_K$.

$$\begin{aligned} \gamma &= \sum_i c_i \omega_i \\ |N\gamma| &= \left| \prod_j \left(\sum_i c_i \omega_i^{(j)} \right) \right| \leq \prod_j \sum_i |c_i| \cdot |\omega_i^{(j)}| \leq \\ &\leq C(\max |c_i|)^n, \end{aligned}$$

where $\omega_i^{(j)}$ denotes the conjugates of ω_i and

$$C = \prod_j \sum_i |\omega_i^{(j)}|.$$

Take $m \in \mathbb{Z}, m > \sqrt[n]{C}, M := m^n$. We now define the integral and fractional part of γ as follows

$$\begin{aligned} \lfloor \gamma \rfloor &:= \sum_i \lfloor c_i \rfloor \omega_i \in \mathcal{O}_K \\ \{ \gamma \} &:= \sum_i \{ c_i \} \omega_i. \end{aligned}$$

Now $\gamma = \lfloor \gamma \rfloor + \{ \gamma \}$ and consider the injective map

$$\begin{aligned} \varphi : K &\hookrightarrow \mathbb{R}^n \\ \sum c_i \omega_i &\mapsto (c_1, \dots, c_n) \\ \varphi(\{ \gamma \}) &\in [0, 1)^n. \end{aligned}$$

We now divide the unit cube $[0, 1]^n$ into $m^n = M$ smaller cubes of side $\frac{1}{m}$ and consider the points $\varphi(\{k\gamma\})$ for $1 \leq k \leq M+1$. Then there exist two such points that lie in the same smaller cube, say $\ell\gamma, k\gamma, k < \ell, 0 < t := \ell - k \leq M$. We then have that $t\gamma = \omega + \varepsilon$ for $\omega \in \mathcal{O}_K, \varepsilon \in K$ such that all coordinates of $\varphi(\varepsilon)$ lie in $(-\frac{1}{m}, \frac{1}{m})$. Thus

$$|N\varepsilon| < C \left(\frac{1}{m} \right)^n < 1.$$

□

Theorem 1.14. *Class number is always finite.*

Proof. Let $A < \mathcal{O}_K, \beta \in A, \beta \neq 0$ such that $|N(\beta)|$ is minimal. For $\alpha \in A \exists t; 1 \leq t \leq M$ such that

$$|N(t\alpha - \omega\beta)| < |N(\beta)|.$$

The minimality of $|N(\beta)|$ implies that $t\alpha = \omega\beta \in (\beta)$. Now

$$t\alpha \mid M!\alpha \implies (\beta) \supset (t\alpha) \supset M!(\alpha).$$

Hence $M!A \subset (\beta)$. Let $B := \frac{1}{\beta}M!A < \mathcal{O}_K$. We have $(M!)A = (\beta)B$ and thus $A \sim B$. Furthermore we have

$$\beta \in A \implies M!\beta \in (\beta)B = M!A \implies M! \in B, (M!) \subset B.$$

However since $\mathcal{O}_K/(M!)$ is finite by Prop. 1.9 and B corresponds to an ideal in $\mathcal{O}_K/(M!)$, then we only have finitely many ideals containing $M!$, hence finitely many, that can be equivalent with A , which concludes the proof. $\square$

Corollary 1.15. $\forall A < \mathcal{O}_K \exists k : 1 \leq k \leq h_K$ such that A^k is principal.

Proof. Consider the set $\{A^i \mid 1 \leq i \leq h_K + 1\}$. At least two of the ideals in this set are equivalent. Let $A^i \sim A^j$ where $i < j$ and let $\alpha, \beta \in \mathcal{O}_K$ such that $(\alpha)A^i = (\beta)A^j$. We set $k := j - i$ and $B := A^k$ and show that B is principal. We have

$$(\alpha)A^i = (\beta)BA^i \implies \left(\frac{\alpha}{\beta}\right)A^i = BA^i \subset A \xrightarrow{1.11} \frac{\alpha}{\beta} \in \mathcal{O}_K.$$

By Prop. 1.12 $(\frac{\alpha}{\beta}) = B$ is principal. $\square$

Corollary (Ideal classes form a group). *For $A < \mathcal{O}_K$ we denote by $[A]$ its ideal class. We can equip the ideal classes with group structure as follows:*

We define the group operation as

$$[A] \cdot [B] = [AB]$$

with the unit element $[\mathcal{O}_K]$ and the inverse element

$$[A]^{-1} := [A^{k-1}]$$

where k is such that A^k is principal (such k exists by Cor. 1.15). The resulting group is denoted by $\mathcal{C}\ell_K$.

1.6 Cancellation and unique factorization

Proposition 1.16. *Let $A, B, C < \mathcal{O}_K$, $A \neq 0$.*

(a) If $AB = AC$, then $B = C$.

(b) $B \supset A$ iff $B \mid A$.

Proof. (a) By 1.15 we have k such that $A^k = (\alpha)$.

$$A^k B = A^k C$$

$$(\alpha)B = (\alpha)C$$

$$B = C$$

(b)

$$B^l = (\beta)$$

$$B^{l-1}A \subset \mathcal{O}_K$$

$$C = \frac{1}{\beta}B^{l-1}A < \mathcal{O}_K.$$

Then $BC = A$. $\square$

Theorem 1.17. *Each ideal in $\mathcal{O}_K$ has a unique factorization into a product of prime ideals.*

Proof. Existence: Let A be a proper ideal and let P_1 be a maximal ideal (which is prime by 1.10) containing A . By 1.16 P_1 divides A , therefore there exists an ideal $A_1 < \mathcal{O}_K$ such that $P_1 A_1 = A$. If $A_1 = \mathcal{O}_K$, then we are done. Otherwise A_1 is a proper ideal, so we can iterate this process, obtaining

$$A \subset A_1 \subset A_2 \subset \dots$$

Since $\mathcal{O}_K$ is noetherian, there exists i such that $A_i = \mathcal{O}_K$, then $A = P_1 P_2 \dots P_i$.

Uniqueness: Let $P_1 \dots P_k = Q_1 \dots Q_l$. Thus

$$P_1 | Q_1 \dots Q_l \implies P_1 \supset Q_1 \dots Q_l.$$

WLOG $P_1 \supset Q_1$, then $P_2 \dots P_k = Q_2 \dots Q_l$ and by continuing like this, we arrive at the two factorizations being equal. $\square$

Definition. Let $P < \mathcal{O}_K$ be a prime ideal and $A < \mathcal{O}_K$ an ideal. We denote by $\text{ord}_P A := v_P A$ the unique $t \in \mathbb{Z}_{\geq 0}$ such that $P^t | A$ and $P^{t+1} \nmid A$.

Remark. Here are some basic properties of ord :

- (a) $\text{ord}_P AB = \text{ord}_P A + \text{ord}_P B$,
- (b) $A = \prod P^{\text{ord}_P A}$.

1.7 Ideal group

We have already introduced the notion of ideal class group. In this section we will introduce the notion of ideal group. The set $\{I < \mathcal{O}_K \mid I \neq \{0\}\}$ is a commutative semigroup (or monoid).

We can obtain a group by considering formal fractions AB^{-1} (ordered pairs of ideals with equivalence defined as $AB^{-1} = (AC)(BC)^{-1}$, which is well-defined thanks to the cancellation property). This group is denoted by $\mathcal{I}_K$ and is called the *ideal group* of the field K .

We can view the formal fractions as subsets of K by recalling that there always exists a $k \in \mathbb{N}$ such that B^k is principal (by Cor. 1.15). Let $B^k = (\beta)$. Then we have

$$AB^{-1} = \frac{AB^{k-1}}{(\beta)} = \underbrace{\frac{1}{\beta} AB^{k-1}}_{\subset K} < \mathcal{O}_K.$$

Sets of this form are called *fractional ideals*. Alternatively they can be defined as finitely generated $\mathcal{O}_K$ -submodules in K . As an exercise, you can show that these are equivalent definitions.

For each $I < \mathcal{O}_K$, we have $I = I\mathcal{O}_K^{-1}$ and thus $I \in \mathcal{I}_K$. Thanks to the cancellation property it follows that $I\mathcal{O}_K^{-1} = J\mathcal{O}_K^{-1} \iff I = J$.

Definition. The set $\{(\alpha)(\beta) \mid \alpha, \beta \in \mathcal{O}_K \setminus \{0\}\}$ denoted by $\mathcal{P}_K$ is called the *group of principal fractional ideals* of the field K .

Exercise. Show, that

$$\mathcal{I}_K / \mathcal{P}_K = \mathcal{C}\ell_K.$$

1.8 Prime factorizations

We will now focus more on the prime ideals and how they look like in $\mathcal{O}_K$ for some given number field K .

Let $P < \mathcal{O}_K$ be a prime ideal. $P \cap \mathbb{Z}$ is an ideal in $\mathbb{Z}$ and by Observation 2 we have $P \cap \mathbb{Z} \neq \{0\}$. Since $\mathbb{Z}$ is a PID, then $P \cap \mathbb{Z} = a\mathbb{Z}$ for some $a \in \mathbb{Z}$. It can easily be shown that it in fact must be equal to some $p\mathbb{Z}$, where p is a prime number. This then means that there exists a unique prime number p such that $p \in P$. We would now like to factor all such $(p) = p\mathcal{O}_K$.

Definition. Let $P < \mathcal{O}_K$ be a prime ideal and $p \in P$ a prime number. We define the *ramification index* of P as

$$e := e(P/p) := \text{ord}_P(p).$$

We also define the *inertia degree* of P as $f \geq 1$ such that

$$\#\mathcal{O}_K/P = p^f$$

and denote it by f or $f(P/p)$.

Lemma. $\#\mathcal{O}_K/(p) = p^n$, where $n = [K : \mathbb{Q}]$.

Proof. See 1.9. □

Theorem 1.18 (EFG theorem). *Let $p \in \mathbb{Z}$ be a prime number, $P_1, \dots, P_g < \mathcal{O}_K$ prime ideals containing p . Let e_i, f_i be their ramification indices and inertia degrees. Then*

$$\sum_{i=1}^g e_i f_i = n = [K : \mathbb{Q}].$$

Moreover if $K/\mathbb{Q}$ is a Galois extension, then $e_1, \dots, e_g =: e, f_1, \dots, f_g =: f$ and $efg = n$.

Remark. The EFG theorem works for general field extensions L/K as well. However this weaker version will be sufficient for us.

Definition. Let $p \in \mathbb{Z}$ be a prime number and P_i, e_i, f_i, g as in the EFG theorem. We say, that
(a) p *splits completely* if $g = n, e_i = f_i = 1$ for all i and

$$(p) = P_1 \dots P_n.$$

(b) p is *inert* if $g = 1, e_1 = 1, f_1 = n$ and

$$(p) = P.$$

(c) p *ramifies* if there exists an index i such that $e_i \geq 2$.

Remark. Later on we will prove that there are only finitely many prime numbers that ramify and they can be characterized.

Proposition 1.19. *Let $P < \mathcal{O}_K$ be a prime ideal of inertia degree f . Then*

$$\#\mathcal{O}_K/P^k = p^{kf} \quad k \in \mathbb{N}.$$

Proof. The proof is done by induction.

For $k = 1$ it follows from definition.

Now let $k > 1$. We will view $\mathcal{O}_K/P^k$ as a group. Then it has a subgroup P^{k-1}/P^k . By 2nd isomorphism theorem

$$(\mathcal{O}_K/P^k)/(P^{k-1}/P^k) \simeq \mathcal{O}_K/P^{k-1}.$$

By inductive hypothesis, we have

$$\#\mathcal{O}_K/P^{k-1} = p^{(k-1)f}.$$

Now we will show that

$$\#P^{k-1}/P^k = p^f.$$

Because factorization into prime ideals is unique, we have $P^{k-1} \neq P^k$, so we can choose $\alpha \in P^{k-1} \setminus P^k$. By 1.16 we have $(\alpha) + P^k \mid P^k$. Again by 1.17 it follows that

$$(\alpha) + P^k = P^l.$$

for some l . We also have

$$P^{k-1} \supset (\alpha) + P^k \implies P^{k-1} \mid (\alpha) + P^k.$$

Hence $(\alpha) + P^k$ is either P^{k-1} or P^k . Since $\alpha \notin P^k$, then $(\alpha) + P^k = P^{k-1}$. Consider now the surjection

$$\begin{aligned} \varphi: \mathcal{O}_K &\rightarrow P^{k-1}/P^k \\ \gamma &\mapsto \gamma\alpha + P^k \end{aligned}$$

which is also a group homomorphism. Let us now inspect its kernel. We have

$$\begin{aligned} \gamma \in \text{Ker } \varphi &\iff \gamma\alpha \in P^k \iff \text{ord}_P(\gamma\alpha) \geq k \\ k &\leq \text{ord}_P(\gamma\alpha) = \text{ord}_P(\gamma) + \text{ord}_P(\alpha) = \text{ord}_P(\gamma) + k - 1 \\ \gamma \in \text{Ker } \varphi &\iff \text{ord}_P(\gamma) \geq 1 \iff \gamma \in P \end{aligned}$$

Hence $\text{Ker } \varphi = P$. By 1st isomorphism theorem

$$\mathcal{O}_K/P \simeq P^{k-1}/P^k$$

which concludes the proof. $\square$

Recall. Recall the Chinese remainder theorem (CRT for short), proof of which can be found in ÚKA under Prop. 1.21.

Let R be a unital commutative ring and $A_1, \dots, A_g < R$ pairwise comaximal ideals i.e. $A_i + A_j = R$ for $i \neq j$. Then

$$\begin{aligned} R/A_1 \dots A_g &\simeq R/A_1 \times \dots \times R/A_g \\ A_1 \dots A_g &= A_1 \cap \dots \cap A_g \end{aligned}$$

Proof of Thm. 1.18, part 1. Let $(p) = P_1^{e_1} \dots P_g^{e_g}$. As an exercise, show that these are pairwise comaximal. By CRT

$$\mathcal{O}_K/(p) \simeq \mathcal{O}_K/P_1^{e_1} \times \dots \times \mathcal{O}_K/P_g^{e_g}.$$

The size of LHS is

$$\#\mathcal{O}_K/(p) = p^n.$$

By 1.19 we have

$$\#RHS = \prod p^{e_i f_i} = p^{\sum e_i f_i}.$$

From the isomorphism it now follows that $n = \sum e_i f_i$. $\square$

Definition. Let $K/\mathbb{Q}$ be a Galois extension, $A < \mathcal{O}_K$ an ideal and $\sigma \in \text{Gal}(K/\mathbb{Q})$. Then we define

$$\sigma A := \{\sigma(\alpha) \mid \alpha \in A\} < \mathcal{O}_K.$$

Observation 3. Observe that the following holds

- (a) $\sigma \mathcal{O}_K = \mathcal{O}_K \implies \mathcal{O}_K / \sigma A = \sigma \mathcal{O}_K / \sigma A \simeq \mathcal{O}_K / A$.
- (b) For a prime ideal P , σP is also a prime ideal.

Proposition 1.20. *Let $p \in \mathbb{Z}$ be a prime number, $P, Q < \mathcal{O}_K$ prime ideals dividing (p) . If $K/\mathbb{Q}$ is Galois, then there exists $\sigma \in \text{Gal}(K/\mathbb{Q})$ such that $\sigma P = Q$ i.e. the Galois group acts transitively on $\{P_1, \dots, P_g\}$.*

Proof. For the purpose of this proof, let us denote $G := \text{Gal}(K/\mathbb{Q})$. Assume towards contradiction that $Q \notin \{\sigma P \mid \sigma \in G\}$. By CRT there exists $\alpha \in \mathcal{O}_K$ such that

$$\begin{aligned} \alpha &\equiv 0 \pmod{Q} \\ \alpha &\equiv 1 \pmod{\sigma P} \quad \forall \sigma \in G. \end{aligned}$$

Now we have

$$N(\alpha) = \prod \sigma \alpha \in Q \cap \mathbb{Z} = (p).$$

Then necessarily

$$P \mid (p) \mid (N(\alpha)) = \prod (\sigma \alpha) \implies P \mid (\sigma \alpha)$$

for some σ . Thus we have $\sigma \alpha \in P$, which is a contradiction with $\alpha \equiv 1 \pmod{\sigma^{-1}P}$. $\square$

The second part of Thm. 1.18 follows from Prop. 1.20.

1.9 Finding factors

Theorem 1.21. *Let $K/\mathbb{Q}$ be a field extension such that $\mathcal{O}_K = \mathbb{Z}[\alpha]$ for some α . Let $f \in \mathbb{Z}[x]$ denote the minimal polynomial for α and p be a prime number. For $g \in \mathbb{Z}[x]$ we shall denote $\bar{g} \in \mathbb{F}_p[x]$ and let*

$$\bar{f} = \prod_{i=1}^r \bar{g}_i(x)^{e_i}$$

be the factorization into irreducibles over $\mathbb{F}_p$, where g_i 's are monic polynomials in $\mathbb{Z}[x]$. Then

$$p\mathcal{O}_K = \prod_{i=1}^r (p, g_i(\alpha))^{e_i}$$

is the distinct prime ideal factorization. Moreover

$$\mathcal{O}_K / (p, g_i(\alpha)) \simeq \mathbb{F}_p[x] / \bar{g}_i$$

thus

$$f_i = f = \left((p, g_i(\alpha)) / p \right) = \deg g_i.$$

Proof. Consider the isomorphism

$$\begin{aligned} \mathbb{Z}[x] / (f(x)) &\simeq \mathcal{O}_K = \mathbb{Z}[\alpha] \\ x + (f(x)) &\mapsto \alpha. \end{aligned}$$

Factoring mod p gives us

$$\begin{aligned}\mathbb{F}_p[x]/(\bar{f}(x)) &\simeq \mathcal{O}_K/(p) \\ x + (\bar{f}(x)) &\mapsto \alpha + (p).\end{aligned}$$

Now $\mathbb{F}_p[x]/(\bar{f})$ has maximal ideals $(\bar{g}_i) := (\bar{g}_i + (\bar{f}))$ for $i \in \{1, \dots, r\}$. Now we have

$$\prod (\bar{g}_i)^{e_i} = 0 \equiv \bar{f}$$

and let us show that no smaller e_i 's give 0.

The ideal $(\bar{g}_i)$ corresponds to $(g_i(\alpha) + (p)) < \mathcal{O}_K/(p)$ and this by the 2nd isomorphism theorem corresponds to $P_i := (p, g_i(\alpha)) < \mathcal{O}_K$. These P_i are all the prime ideals in $\mathcal{O}_K$ such that they contain (p) . If we have

$$p\mathcal{O}_K = \prod P_i^{e'_i},$$

then the e'_i are characterized by the property

$$p\mathcal{O}_K \supset \prod P_i^{e'_i},$$

but they do not contain any product with smaller exponents. Under the correspondence, it corresponds to $\prod (\bar{g}_i)^{e_i}$, hence $e_i = e'_i$, which concludes the proof. $\square$

Example. Consider the number field $K = \mathbb{Q}[\sqrt{-5}]$. Then $\mathcal{O}_K = \mathbb{Z}[\sqrt{-5}]$ and $f = x^2 + 5$ is the minimal polynomial for $\sqrt{-5}$.

For $p = 2$ we have

$$\begin{aligned}x^2 + 5 &\equiv (x + 1)^2 \pmod{2} \\ r = 1 \quad g_1 = x + 1 \in \mathbb{Z}[x] \quad e_1 = 2 \\ 2\mathcal{O}_K = (2) &= (2, \sqrt{-5} + 1)^2 \implies 2 \text{ ramifies}\end{aligned}$$

For $p = 3$ we have

$$\begin{aligned}x^2 + 5 &\equiv (x + 1)(x - 1) \pmod{3} \\ 3\mathcal{O}_K = (3) &= (3, \sqrt{-5} + 1)(3, \sqrt{-5} - 1) \implies 3 \text{ splits completely}\end{aligned}$$

For $p = 11$, $x^2 + 5$ is irreducible mod 11

$$11\mathcal{O}_K = (11) = (11, (\sqrt{-5})^2 + 5) = (11) \implies 11 \text{ is inert.}$$

Exercise. (a) In the example above, characterize all p . (Hint: Try using Legendre symbols)

(b) Characterize all p for general number field $\mathbb{Q}[\sqrt{D}]$.

(c) Use 1.21 to prove 1.18.

1.10 Ramification

Theorem 1.22. *Let p be a prime number. Then p ramifies in K if and only if $p \mid \Delta_K$.*

In order to prove this theorem, we first need to introduce a generalization to the discriminant. We assume $A \subset B$ to be rings such that B is a free A -module of rank m i.e.

$$B \simeq A^m$$

as an A -module. Now for an element $\beta \in B$ we can define a map

$$\begin{aligned} B &\rightarrow B \\ x &\mapsto \beta x \end{aligned}$$

which can be viewed as a linear map $A^m \rightarrow A^m$. Then we can define $\text{Tr}_{B/A} \beta$ as the trace of this linear map. More concretely for an A -basis $e_1, \dots, e_m$ we have

$$\begin{aligned} \beta e_j &= \sum a_{ij} e_i \\ \text{Tr}_{B/A} \beta &= \sum a_{ii} \in A. \end{aligned}$$

Now given some $\beta_1, \dots, \beta_m \in B$ we define

$$\begin{aligned} \Delta(\beta_1, \dots, \beta_m) &:= \det(\text{Tr}_{B/A}(\beta_i \beta_j)) \in A \\ \Delta(B/A) &:= (\Delta(e_1, \dots, e_m)) \in A. \end{aligned}$$

Lemma 1.23. *Let $B \supset A$ with a basis $e_1, \dots, e_m$ and $I < A$ be an ideal. Then*

$$\bar{e}_i := e_i + IB \quad i \in \{1, \dots, m\}$$

is a basis for B/IB as an A/I -module and

$$\Delta(\bar{e}_1, \dots, \bar{e}_m) \equiv \Delta(e_1, \dots, e_m) \pmod{I}.$$

Lemma 1.24.

$$\Delta(\Pi B_i/A) = \prod \Delta(B_i/A)$$

Definition. Let R be a ring and $\alpha \in R$. We say that α is *nilpotent* if $\alpha^r = 0$ for some $r \geq 1$.

Proof of Thm 1.22. By CRT we have

$$\mathcal{O}_K/(p) \simeq \mathcal{O}_K/P_1^{e_1} \times \dots \times \mathcal{O}_K/P_g^{e_g}.$$

Observation. $\mathcal{O}_K/P_i^{e_i}$ contains a non-zero nilpotent element iff $e_i \geq 2$.

Proof of the Observation. If $e_i = 1$, then since P_i is prime and maximal ideal, then $\mathcal{O}_K/P_i$ is a field and so does not contain any nilpotent elements.

Now if $e_i \geq 2$, then $\alpha \in P_i \setminus P_i^{e_i}$ is nilpotent. □

Now from the observation above it also trivially follows that

$$\mathcal{O}_K/P_1^{e_1} \times \dots \times \mathcal{O}_K/P_g^{e_g}$$

contains a non-zero nilpotent element iff $e_i \geq 2$ for some i . In other words it is sufficient for one of the factor rings to contain a nilpotent element.

We have

$$\begin{aligned} \Delta_K &= \Delta(\mathcal{O}_K/\mathbb{Z}) \\ &\equiv \Delta\left(\mathcal{O}_K/(p)/\mathbb{F}_p\right) \pmod{p} \\ &= \prod \Delta\left(\mathcal{O}_K/P_i^{e_i}/\mathbb{F}_p\right) \pmod{p}. \end{aligned}$$

If $e_i = 1$, then $\mathcal{O}_K/P_i$ is a field and $(\mathcal{O}_K/P_i)/\mathbb{F}_p$ is a separable field extension by ÚKA Cor. 2.18, hence

$$\Delta(\mathcal{O}_K/P_i/\mathbb{F}_p) \neq 0.$$

If all $e_i = 1$, then

$$\prod \Delta(\mathcal{O}_K/P_i^{e_i}/\mathbb{F}_p) \not\equiv 0 \pmod{p} \implies p \nmid \Delta_K.$$

If $e_i \geq 2$ for some i , then $\mathcal{O}_K/(p)$ has a nilpotent element $\beta \neq 0$. We choose a $\mathbb{F}_p$ -basis for $\mathcal{O}_K/(p)$

$$\beta = b_1, \dots, b_f.$$

One can check that $\text{Tr}(\beta b_i) = 0$ for all i , hence

$$\Delta(\mathcal{O}_K/(p)/\mathbb{F}_p) = \det(\text{Tr}(b_i b_j)) = 0 \text{ in } \mathbb{F}_p,$$

hence $p \mid \Delta_K$. □

2. Geometry of numbers

2.1 Ideal norm

Definition. For an ideal $I < \mathcal{O}_K$, we define its (*numerical*) *norm* as $\mathcal{N}I = 0$ if $I = (0)$ and

$$\mathcal{N}I = \#\mathcal{O}_K/I$$

otherwise.

Observation. (a) By Prop. 1.9 the norm $\mathcal{N}I$ is always finite and lies in $\mathbb{Z}^+$.

(b) From the proof of Prop. 1.9 it follows that

$$\mathcal{N}(a\mathcal{O}_K) = |a|^n \quad a \in \mathbb{Z}.$$

(c) $I \mid \mathcal{N}I$.

Proposition 2.1. (a) Let $P < \mathcal{O}_K$ be a prime ideal and p a prime number contained in P , then

$$\mathcal{N}P = p^{f(P/(p))}.$$

(b) Let $I = \prod P_i^{r_i}$ for P_i prime ideals and $r_i \geq 1$. Then

$$\mathcal{N}I = \prod p_i^{f_i r_i},$$

where p_i is a prime number contained in P_i .

(c) $\mathcal{N}(IJ) = \mathcal{N}(I) \cdot \mathcal{N}(J)$.

(d) If $I \supset J$, then

$$\#I/J = \mathcal{N}(I^{-1}J).$$

Proof. (a) Follows from definition of the inertia degree f .

(b) By CRT we have

$$\mathcal{O}_K/I \simeq \mathcal{O}_K/P_1^{r_1} \times \cdots \times \mathcal{O}_K/P_m^{r_m}.$$

By Prop. 1.19

$$\#\mathcal{O}_K/P_i^{r_i} = p_i^{r_i f_i},$$

which concludes part (b).

(c) Factor I, J into prime ideals. Then the rest follows from part (b).

(d) By 2nd isomorphism theorem

$$\begin{aligned} (\mathcal{O}_K/J)/(I/J) &\simeq \mathcal{O}_K/I \\ \#\mathcal{O}_K/J &= \#I/J \cdot \#\mathcal{O}_K/I \end{aligned}$$

and the rest is left as an exercise. □

Proposition 2.2. (a) Given a Galois extension $K/\mathbb{Q}$, prime number p and prime ideal $P < \mathcal{O}_K$ such that $P \mid (p)$, let

$$p\mathcal{O}_K = (P_1 \dots P_g)^e.$$

Then

$$\mathcal{N}P\mathcal{O}_K \stackrel{(1)}{=} (P_1 \dots P_g)^{ef} \stackrel{(2)}{=} \prod_{\sigma \in \text{Gal}(K/\mathbb{Q})} \sigma P.$$

(b) For $\beta \in \mathcal{O}_K$, we have $|N(\beta)| = \mathcal{N}(\beta\mathcal{O}_K)$.

Proof of part (a). We have

$$\mathcal{N}P\mathcal{O}_K = p^f \mathcal{O}_K = (p\mathcal{O}_K)^f = (P_1 \dots P_g)^{ef},$$

which concludes the equality (1).

By Prop. 1.20 the Galois group $G := \text{Gal}(K/\mathbb{Q})$ acts transitively on $\{P_1, \dots, P_g\}$ and therefore the action has only one orbit of size g . This means that for all i the stabilizer

$$G_{P_i} = \{\sigma \in G \mid \sigma P_i = P_i\}$$

has the size equal to

$$\frac{\#G}{\#\text{orbit of } P_i} = \frac{n}{g} \stackrel{1.18}{=} ef.$$

The rest of the equality (2) is left as an exercise. The proof of part (b) will be provided later. $\square$

Remark. Part (b) implies that the following diagram commutes

$$\begin{array}{ccc} K^\times & \xrightarrow{\beta \mapsto (\beta)} & \mathcal{I}_K \\ \downarrow N_{K/\mathbb{Q}} & & \downarrow \mathcal{N}: \frac{I}{J} \mapsto \frac{\mathcal{I}}{\mathcal{J}} \\ \mathbb{Q}^\times & \xrightarrow{|\cdot|} & \mathbb{Q}^+ (\simeq \mathcal{I}_{\mathbb{Q}}) \end{array}$$

2.2 Minkowski bound

Theorem 2.3 (Minkowski bound). *Let $K/\mathbb{Q}$ be a field extension of degree n and let $2s$ denote the number of embeddings $K \hookrightarrow \mathbb{C}$, whose image does not lie in $\mathbb{R}$. Then there exists a set of representatives for the class group $\mathcal{C}\ell_K$ consisting of ideals $I < \mathcal{O}_K$ with the property*

$$\mathcal{N}I \leq \frac{n!}{n^n} \left(\frac{4}{\pi} \right)^s |\Delta_K|^{\frac{1}{2}}.$$

Remark. (a) If we let $\mathbb{Q}(\alpha) = K$, then the number $2s$ is the number of roots of m_α , which lie in $\mathbb{C} \setminus \mathbb{R}$.

(b) The term $\frac{n!}{n^n} \left(\frac{4}{\pi} \right)^s$ is called the Minkowski constant. The whole RHS is called the Minkowski bound and is often denoted by B_K .

(c) We often denote by r the number of real roots of m_α , which then gives the equality $n = 2s + r$.

Corollary 2.4. *There is no unramified field extension $K/\mathbb{Q}$ i.e. in each number field K there exists a prime number p that ramifies in K .*

Proof. Since $h_K \geq 1$ for any field K , there exists an ideal $I < \mathcal{O}_K$ from Thm. 2.3 such that

$$1 \leq \mathcal{N}I \leq \frac{n!}{n^n} \left(\frac{4}{\pi} \right)^s |\Delta_K|^{\frac{1}{2}}.$$

After rearranging we obtain

$$|\Delta_K|^{\frac{1}{2}} \geq \frac{n^n}{n!} \left(\frac{\pi}{4}\right)^s > 1,$$

where the last inequality is an analysis exercise. Now $|\Delta_K| \geq 2$ and so there exists a prime number p such that $p \mid \Delta_K$, which ramifies by Thm. 1.22. $\square$

Corollary 2.5. *The class number h_K is finite.*

Proof. Take an ideal $I < \mathcal{O}_K$ and factor it into prime ideals

$$I = \prod P_i^{r_i} \implies B_K \geq \mathcal{N}I = \prod (\mathcal{N}P_i)^{r_i} \implies 2 \leq \mathcal{N}P_i \leq B_K.$$

Now there are only finitely many candidates for P_i for bounded r_i , hence h_K is finite. $\square$

Finding the class group $\mathcal{C}\ell_K$:

- (a) Find $S = \{I < \mathcal{O}_K \mid \mathcal{N}I \leq B_K\}$.
- (b) For $I, J \in S$ test, whether $I \sim J \pmod{P_K}$, where P_K is principal.
- (c) Find relations $[IJ] = [H]$ and $[I^{-1}] = [J]$.

For (b) and (c) it suffices to test when $I \sim J$ i.e. whether IJ^{-1} is principal.

There exists $H < \mathcal{O}_K$ such that $JH = (\mathcal{N}J)$, then

$$IJ^{-1} = \frac{IH}{JH} = \frac{IH}{(\mathcal{N}J)}.$$

It now suffices to see if $IH = (\mathcal{N}J)(\alpha) =: (\beta)$ for some $\alpha \in \mathcal{O}_K$ as then IJ^{-1} is principal. We have

$$\mathcal{N}(IH) = \mathcal{N}(\beta) \stackrel{2.2}{=} |N\beta|,$$

which leads to a diophantine equation.

Example. Consider the case $K := \mathbb{Q}(\sqrt{-5})$. Then $\mathcal{O}_K = \mathbb{Z}[\sqrt{-5}]$ and consider the ideal

$$(3) = (3, \sqrt{-5} - 1)(3, \sqrt{-5} + 1).$$

Does there exist α such that

$$N\alpha = \mathcal{N}(3, \sqrt{-5} - 1) = 3?$$

Rewriting $N\alpha$ as $a^2 + 5b^2$ yields a diophantine equation

$$a^2 + 5b^2 = 3,$$

which clearly has no solutions, hence the ideal (3) is not principal in $\mathcal{O}_K$.

Example. For $K := \mathbb{Q}(i)$ and an ideal $I < \mathcal{O}_K$, we can write the Minkowski bound

$$\mathcal{N}(I) \leq B_K = 1.27\dots,$$

hence $I = \mathcal{O}_K$ for all ideals. Therefore the class group $\mathcal{C}\ell_K$ has only one element, hence $\mathbb{Q}(i)$ is a PID.

2.3 Lattices

Definition. Let V be a real vector space of dimension $n < \infty$. A *lattice* Λ in V is a (additive) subgroup

$$\Lambda = \mathbb{Z}e_1 + \cdots + \mathbb{Z}e_r,$$

where e_i 's are linearly independent over $\mathbb{R}$.

A lattice is said to be *full* if $r = n$.

Example. If we identify $\mathbb{Z}[i] \simeq \mathbb{Z}^2$, then we have $\mathbb{Z}^2 \subset \mathbb{R}^2$, hence $\mathbb{Z}[i]$ is a lattice.

On the other hand, consider $\mathbb{Z}[\sqrt{2}] = \mathbb{Z} \cdot 1 + \mathbb{Z}\sqrt{2}$. Since 1 and $\sqrt{2}$ are not linearly independent over $\mathbb{R}$, then $\mathbb{Z}[\sqrt{2}]$ is not a lattice.

Lemma 2.6. *Let V be a finite-dimensional real vector space and $\Lambda < V$ an additive subgroup. Then Λ is a lattice iff it is discrete i.e.*

$$\forall \lambda \in \Lambda \exists U \subset V \text{ open such that } U \cap \Lambda = \{\lambda\}.$$

Proof. Proof is left as an exercise. □

Definition. Let $\Lambda = \sum \mathbb{Z}e_i$ be a full lattice. We call the set

$$D := \left\{ \sum a_i e_i \mid 0 \leq a_i < 1 \right\}$$

the *fundamental parallelepiped* in Λ .

Definition. By *volume* we shall call a measure on V and denote it by Vol . Let Λ be a lattice with the fundamental parallelepiped D . Its *covolume* is the value $\text{Vol } D$.

Observation 4. (a)

$$V = \bigsqcup_{\lambda \in \Lambda} (\lambda + D).$$

(b)

$$\forall v \in V \exists! \lambda \in \Lambda : v - \lambda \in D.$$

In other words D is a fundamental domain of the action of shifting by Λ .

(c) From linear algebra, we know

$$\text{Vol}(D) = |\det(e_1, \dots, e_n)|.$$

Moreover the volume is preserved under the change of basis, because changing the basis results in multiplying $\text{Vol}(D)$ by the determinant of the transition matrix, which is 1.

(d) Consider full lattices $\Lambda' \subset \Lambda \subset V$. Then

$$\frac{\text{Vol}(D')}{\text{Vol}(D)} = (\Lambda : \Lambda'),$$

where $(\Lambda : \Lambda')$ is the index of a subgroup.

Proof. (d) There exists a basis $e_1, \dots, e_n$ for Λ and numbers $m_1, \dots, m_n \in \mathbb{Z}^+$ such that $m_i e_i$ is a basis for Λ' . Then $(\Lambda : \Lambda') = \prod m_i$. You can then view the LHS as cutting D' into $m_1, \dots, m_n$ pieces in the corresponding directions given by e_i , which yields the desired equality.

Why does such a basis exist? Take $\Lambda = \sum \mathbb{Z}e_i$ and let f_j be a basis for Λ' . The matrix consisting of the coordinates of f_i 's with respect to e_i when diagonalized, yields

$$\begin{pmatrix} m_1 & & 0 \\ & \ddots & \\ 0 & & m_n \end{pmatrix}$$

□

Remark. Based on part (b) of the observation above, we shall use the terms "fundamental domain" and "fundamental parallelepiped" interchangeably in the context of lattices.

Proof of Prop. 2.2, part (b). By definition we have

$$\mathcal{N}(\beta\mathcal{O}_K) = \#\mathcal{O}_K/\beta\mathcal{O}_K.$$

Let $\omega_1, \dots, \omega_n$ be an integral basis for $\mathcal{O}_K$, then $\beta\omega_1, \dots, \beta\omega_n$ is a basis for $\beta\mathcal{O}_K$ and

$$\mathcal{O}_K = \sum \mathbb{Z}\omega_i \quad \beta\mathcal{O}_K = \sum \mathbb{Z}\beta\omega_i,$$

hence we can view $\beta\mathcal{O}_K \subset \mathcal{O}_K \subset \mathbb{R}^n$ as lattices. Then $\mathcal{O}_K$ corresponds to Λ in Observation 4 part (d) and $\beta\mathcal{O}_K$ corresponds to Λ' . In this context, $\mathcal{N}(\beta\mathcal{O}_K)$ corresponds to the subgroup index $(\Lambda : \Lambda')$. □

2.4 Lattice points in convex bodies

Proposition 2.7. *Let $\Lambda \subset V$ be a full lattice with fundamental domain D and let $S \subset V$ be a measurable set. If $\text{Vol } S > \text{Vol } D$, then there exist $\alpha, \beta \in S$ such that*

$$\beta - \alpha \in \Lambda.$$

Proof. For $\lambda \in \Lambda$ we denote $S_\lambda := S \cap (\lambda + D)$. Now we can express the volume of S as follows

$$\text{Vol } S = \sum_{\lambda \in \Lambda} \text{Vol } S_\lambda.$$

The volume remains unchanged under shifting action, hence

$$\text{Vol } S = \sum_{\lambda \in \Lambda} \text{Vol}(\underbrace{-\lambda + S_\lambda}_{\subset D}) > \text{Vol } D,$$

therefore two such sets must intersect. Then there exist $\alpha, \beta \in S$ and $\lambda, \lambda' \in \Lambda$ such that

$$-\lambda + \alpha = -\lambda' + \beta \implies \alpha - \beta = \lambda - \lambda' \in \Lambda.$$

□

Theorem 2.8 (Minkowski). *Let $T \subset V$ be a compact convex subset symmetric around the origin i.e. $\alpha \in T$ iff $-\alpha \in T$. Let Λ be a full lattice on V with the fundamental domain D . If $\text{Vol } T \geq 2^n \text{Vol } D$, then $T \cap \Lambda \neq \{0\}$.*

Proof. Whenever $T \neq \emptyset$, then $0 \in T$, so it is natural for us to ask whether some non-zero element lies in T . It suffices to prove the case with strict inequality i.e.

$$\text{Vol } T > 2^n \text{Vol } D.$$

We set

$$S := \frac{1}{2}T = \left\{ \frac{t}{2} \mid t \in T \right\}.$$

Now $\text{Vol } S > \text{Vol } D$. By Prop. 2.7 there exist $\alpha, \beta = \frac{\gamma}{2}, \frac{\delta}{2} \in S$ such that

$$\Lambda \ni \beta - \alpha = \frac{\delta + (-\gamma)}{2}.$$

$-\gamma \in T$ by symmetry and the whole RHS then lies in T by convexity. □

Corollary 2.9. *Every positive integer is a sum of four squares.*

Proof. It suffices to prove the statement for prime numbers.

For $p = 2$, we have $p = 1 + 1 + 0 + 0$. Let $p > 2$ be a prime number.

Observation 5. The following modular equality

$$m^2 + n^2 + 1 \equiv 0 \pmod{p}$$

has a solution in $\mathbb{Z}$.

Proof of Observation 5. Rewriting the equality as follows

$$m^2 \equiv -n^2 - 1,$$

one can see that there are $\frac{p+2}{2}$ possible values for the RHS and the same number of possible values for the LHS. By pigeonhole principle, at least two must meet and give a solution. □

Fix a solution (m, n) for the equality from the observation above and consider the set $\Lambda \subset \mathbb{Z}^4$

$$\Lambda := \{(a, b, c, d) \mid c \equiv ma + nb, d \equiv mb - na \pmod{p}\}.$$

Since $\mathbb{Z}^4$ is discrete, Λ is also discrete and moreover it is a subgroup of $\mathbb{Z}^4$, hence a lattice. We have

$$a^2 + b^2 + c^2 + d^2 \equiv a^2(1 + m^2 + n^2) + b^2(1 + m^2 + n^2) + 2mnab - 2mnab \equiv 0,$$

therefore multiples of p can be expressed as a sum of 4 squares.

Observation 6. $(\mathbb{Z}^4 : \Lambda) = p^2 \implies \text{Vol } D_\Lambda = p^2$.

Proof of Observation 6. We have $\mathbb{Z}^4 \supset \Lambda \supset (p\mathbb{Z})^4$, hence Λ is full. By second isomorphism theorem

$$\mathbb{Z}^4 / \Lambda \simeq (\mathbb{Z}/p\mathbb{Z})^4 / \Lambda / (p\mathbb{Z})^4.$$

From this, we see that $(\mathbb{Z}^4 : \Lambda) = (\mathbb{F}_p^4 : \Lambda / (p\mathbb{Z})^4)$. The quotient $\Lambda / (p\mathbb{Z})^4$ is a $\mathbb{F}_p$ -vector space of dimension 2, hence the index

$$(\mathbb{F}_p^4 : \Lambda / (p\mathbb{Z})^4) = \frac{\#\mathbb{F}_p^4}{\#\Lambda / (p\mathbb{Z})^4} = \frac{p^4}{p^2} = p^2.$$

□

Let T denote the closed ball of radius r at the origin in $\mathbb{R}^4$ such that $r^2 = (2 - \varepsilon)p$ for some sufficiently small $\varepsilon > 0$.

$$\text{Vol } T = \frac{\pi^2 r^4}{2} = \pi^2 (2 - \varepsilon)^2 p^2 \frac{1}{2} > 2^4 \text{Vol } D_\Lambda,$$

thus by Thm. 2.8 there exists a non-zero $(a, b, c, d) \in T \cap \Lambda$ i.e.

$$\begin{aligned} a^2 + b^2 + c^2 + d^2 &\equiv 0 \pmod{p} \\ a^2 + b^2 + c^2 + d^2 &< r^2 < 2p, \end{aligned}$$

hence $p = a^2 + b^2 + c^2 + d^2$.

Now one checks that the identity

$$\begin{aligned} (a^2 + b^2 + c^2 + d^2)(A^2 + B^2 + C^2 + D^2) &= (aA - bB - cC - dD)^2 + (aB + bA + cD - dC)^2 \\ &\quad + (aC - bD + cA + dB)^2 + (aD + bC - cB + dA)^2 \end{aligned}$$

holds for all positive integers, which concludes the proof. $\square$

Exercise. Show that all prime numbers satisfying $p \equiv 1 \pmod{4}$ can be expressed as a sum of two squares.

2.5 Minkowski space

Definition. Let K be a number field of degree n with r real embeddings $\sigma_1, \dots, \sigma_r : K \hookrightarrow \mathbb{R}$ and $2s$ complex embeddings

$$\sigma_{r+1}, \bar{\sigma}_{r+1}, \dots, \sigma_{r+s}, \bar{\sigma}_{r+s} : K \hookrightarrow \mathbb{C}.$$

We then define the *Minkowski embedding* as the map

$$\begin{aligned} \sigma : K &\hookrightarrow \mathbb{R}^r \times \mathbb{C}^s \\ \alpha &\mapsto (\sigma_1 \alpha, \dots, \sigma_{r+s} \alpha). \end{aligned}$$

We call the real vector space $V := \mathbb{R}^r \times \mathbb{C}^s$ of dimension $r + 2s = n$ the *Minkowski space*.

Proposition 2.10. *Let $I < \mathcal{O}_K$. Then $\sigma(I)$ is a full lattice in V of covolume*

$$\mathcal{N}I \cdot \frac{|\Delta|^{1/2}}{2^s}.$$

Proof. Take an integral basis for the ideal $I = \mathbb{Z}\alpha_1 + \dots + \mathbb{Z}\alpha_n$, such basis exists by Prop. 1.8. The image $\sigma(I)$ is generated by $\sigma(\alpha_1), \dots, \sigma(\alpha_n)$. We need to check whether they are independent over $\mathbb{R}$ to see if they form a lattice.

Let A be a matrix, whose i -th row is of the form

$$(\sigma_1 \alpha_i \dots \sigma_r \alpha_i, \text{Re } \sigma_{r+1} \alpha_i, \text{Im } \sigma_{r+1} \alpha_i, \dots).$$

Matrix A has a non-zero determinant, hence the generators of $\sigma(I)$ are indeed linearly independent over $\mathbb{R}$ and thus $\sigma(I)$ is a lattice.

Now consider a matrix B with i -th row of the form

$$(\sigma_1 \alpha_i \dots \sigma_r \alpha_i, \sigma_{r+1} \alpha_i, \bar{\sigma}_{r+1} \alpha_i, \dots).$$

Matrices A and B are related by some column operations.

$$\det A = (-2i)^{-s} \det B \neq 0.$$

The determinant of B is non-zero, because B is the matrix M from 1.4 and since $\Delta_K \stackrel{1.6}{\neq} 0$, then $\det M \neq 0$. Thus $\sigma(I)$ is a full lattice. Since A can be viewed as having $\sigma\alpha_i$'s as its rows, then $\text{Vol } D = |\det A|$. By 1.6(b)

$$\begin{aligned} |\Delta(\alpha_1, \dots, \alpha_n)| &= (\mathcal{O}_K : I)^2 |\Delta_K| \\ |\det A| &= 2^{-s} (\mathcal{O}_K : I) |\Delta_K|^{1/2} = 2^{-s} |\Delta_K|^{1/2} \mathcal{N}I. \end{aligned}$$

□

Lemma 2.11. *Let $x := (x_1, \dots, x_r, z_{r+1}, \dots, z_{r+s}) \in V$. We define a norm on V as*

$$\|x\| := \sum_{i=1}^r |x_i| + 2 \sum_{i=r+1}^{r+s} |z_i|.$$

Furthermore we shall denote the measure associated with this norm by Meas and define the closed ball in V as the set

$$X(t) := \{x \in V \mid \|x\| \leq t\}.$$

Then

$$\text{Meas } X(t) = \frac{2^r \pi^s t^n}{n! 2^s}.$$

Exercise. Check that Lemma 2.11 holds for $(r, s) = (1, 0), (0, 1)$.

Proposition 2.12. *Let $I < \mathcal{O}_K$. Then there exists a non-zero element $\alpha \in I$ such that*

$$|N\alpha| \leq B_K \cdot \mathcal{N}I.$$

Proof. Let D be the fundamental domain for $\sigma(I)$. The set $X(t)$ is symmetric, convex and compact. Let t be such that we can use Thm. 2.8, by which there exists a non-zero $\sigma(\alpha) \in \sigma(I) \cap X(t)$. Now

$$\begin{aligned} |N\alpha| &= |\sigma_1\alpha| \dots |\sigma_r\alpha| \cdot |\sigma_{r+1}\alpha|^2 \dots |\sigma_{r+s}\alpha|^2 \stackrel{AG}{\leq} \\ &\stackrel{AG}{\leq} \left(\sum_{i=1}^r |\sigma_i\alpha| + 2 \sum_{j=r+1}^{r+s} |\sigma_j\alpha| \right)^n \cdot \frac{1}{n^n} \leq \frac{t^n}{n^n}. \end{aligned}$$

Since we assumed

$$\text{Vol } X(t) \geq 2^n \text{Vol } D,$$

then by 2.10 and 2.11 we have

$$\frac{2^r \pi^s t^n}{n! 2^s} \geq \mathcal{N}I \cdot \frac{|\Delta|^{1/2}}{2^s}.$$

Now choose t such that the expressions above are equal, then we isolate t^n and plug in as follows

$$|N\alpha| \leq \frac{t^n}{n^n} = \frac{n!}{n^n} \cdot \frac{2^n}{2^r \pi^s} \cdot \mathcal{N}I \cdot |\Delta|^{1/2} = B_K \cdot \mathcal{N}I.$$

□

Now we can prove the Minkowski bound from Thm. 2.3

Proof of 2.3. Let $[A] \in \mathcal{C}\ell_K$. Consider the class $[A^{-1}]$, which contains some ideal $J < \mathcal{O}_K$. By 2.12 there exists a non-zero $\beta \in J$ such that

$$|N\beta| \leq B_K \cdot \mathcal{N}J.$$

Since $\beta\mathcal{O}_K \subset J$, then $J \mid \beta\mathcal{O}_K$, thus there exists an ideal $H < \mathcal{O}_K$ such that $HJ = \beta\mathcal{O}_K$. Now

$$[H] = [J^{-1}] = [A] \implies H \in [A].$$

Therefore

$$\begin{aligned} \mathcal{N}J \cdot \mathcal{N}H &= \mathcal{N}(\beta\mathcal{O}_K) \stackrel{2.2}{=} |N\beta| \leq B_K \cdot \mathcal{N}J \\ \mathcal{N}H &\leq B_K. \end{aligned}$$

□

2.6 Dirichlet's unit theorem

In this section we will state the Dirichlet's unit theorem and prove it. The proof will be split into parts, which we will be proving separately throughout this section.

Definition. We denote by $\mu(K)$ the *torsion subgroup* of $\mathcal{O}_K$ i.e. the subgroup of all elements of finite order.

Remark. Each $\zeta \in \mu(K)$ is a root of unity and we will often refer to them as such.

Theorem 2.13 (Dirichlet's unit theorem). *Let K be a number field with r real embeddings and $2s$ complex embeddings. Then $\mathcal{O}_K^\times$ is a finitely generated abelian group of rank $r + s - 1$.*

Definition. The free generators of $\mathcal{O}_K^\times$ are called *the fundamental units* i.e.

$$\mathcal{O}_K^\times = \{\zeta \cdot u_1^{m_1} \cdots u_{r+s-1}^{m_{r+s-1}} \mid \zeta \in \mu(K), m_i \in \mathbb{Z}\},$$

where $\zeta, u_1, \dots, u_{r+s-1}$ are pairwise distinct. The tuple $(u_1, \dots, u_{r+s-1})$ is called the *system of fundamental units*.

Lemma 2.14. *An element $\alpha \in K$ lies in $\mathcal{O}_K^\times$ iff $\alpha \in \mathcal{O}_K$ and $N\alpha = \pm 1$.*

Proof. Let $\alpha \in \mathcal{O}_K^\times$, then it clearly lies in $\mathcal{O}_K$ so it suffices to check the norm condition. Since α is invertible, there exists $\beta \in \mathcal{O}_K^\times$ such that $\alpha\beta = 1$. Then

$$N\alpha N\beta = 1.$$

Since both $N\alpha$ and $N\beta$ lie in $\mathbb{Z}$, then $N\alpha = \pm 1$.

Let $\alpha \in \mathcal{O}_K$ with $N\alpha = \pm 1$.

$$\begin{aligned} \pm 1 &= N\alpha \stackrel{1.2}{=} \prod_{\sigma} \sigma\alpha \\ \pm 1 &= \alpha \cdot \prod_{\sigma \neq \text{id}} \sigma\alpha =: \alpha \cdot \beta \end{aligned}$$

Since K is a field and $\beta = \alpha^{-1}$, then $\beta \in K$. $\sigma\alpha$ are conjugates of α , so they are roots of m_α , hence $\sigma\alpha \in \mathcal{O}_{\mathbb{Q}}$, therefore $\beta \in \mathcal{O}_{\mathbb{Q}}$ and $\beta \in \mathcal{O}_K$. □

Lemma 2.15. *$\mu(K)$ is finite and cyclic.*

Proof. Since $\mu(K)$ is a multiplicative subgroup of a field, then its finiteness implies that it is cyclic, hence it suffices to show that it is finite.

Let $\zeta_m \in \mu(K)$ be an m -th root of unity. Since $\mathbb{Q}(\zeta_m) \subset K$, then $[\mathbb{Q}(\zeta_m) : \mathbb{Q}]$ divides $[K : \mathbb{Q}]$. However there are only finitely many m with $\varphi(m) \leq C$ for each bound C , hence $\mu(K)$ is finite. $\square$

We will now start building up towards the proof of Dirichlet's unit theorem. We will first show that $\mathcal{O}_K^\times$ is finitely generated by proving the following proposition.

Proposition 2.16. *Let $m, M \in \mathbb{Z}$. Then the set of all $\alpha \in \mathcal{O}_{\mathbb{Q}}$ such that*

- $\deg_{m_\alpha} := \deg \alpha \leq m$
- $|\alpha'| \leq M$ for all conjugates α' of α

is finite.

Proof. α is a root of monic $f = x^d + a_{d-1}x^{d-1} + \dots + a_0 \in \mathbb{Z}[x]$ with $\deg f \leq m$. By Viète formula the coefficients of f are also bounded. Since f is monic, we have

$$-a_{d-1} = \sum \alpha'.$$

Now

$$|a_{d-1}| \leq \sum |\alpha'| < dM \leq mM,$$

but then there are only finitely many possible α' . $\square$

Corollary 2.17. *Let $\alpha \in \mathcal{O}_{\mathbb{Q}}$ such that each conjugate α' of α satisfies $|\alpha'| \leq 1$. Then α is a root of unity or zero.*

Proof. Let $k \geq 1$, then $\alpha^k \in \mathbb{Q}(\alpha)$ and hence α^k has a bounded degree. Since taking a conjugate is a homomorphism, we have $|\alpha^k| \leq 1$. By Prop. 2.16 the set $\{1, \alpha, \alpha^2, \dots\}$ is finite, which means $\alpha^k = \alpha^\ell$ for some $k \neq \ell$. Then $\alpha^{\ell-k} = 1$, hence α is either zero or a root of unity. $\square$

Definition. We define a map

$$\begin{aligned} L : K^\times(\cdot) &\hookrightarrow \mathbb{R}^{r+s}(+) \\ \alpha &\mapsto (\log(\sigma_1\alpha), \dots, \log(\sigma_r\alpha), 2\log(\sigma_{r+1}\alpha), \dots), \end{aligned}$$

which is a group homomorphism called the *logarithmic Minkowski embedding*.

Now consider taking $u \in \mathcal{O}_K^\times$, so $|Nu| = 1$. We have

$$\begin{aligned} 1 &= |\sigma_1 u| \cdot \dots \cdot |\sigma_r u| \cdot |\sigma_{r+1} u|^2 \dots \\ 0 &= \log |\sigma_1 u| + \dots + \log |\sigma_r u| + 2 \log |\sigma_{r+1} u| + \dots \end{aligned}$$

The image $L(u)$ thus lies in a hyperplane consisting of solutions to

$$x_1 + \dots + x_{r+s} = 0.$$

We shall denote this hyperplane by H and remark that $H \simeq \mathbb{R}^{r+s-1}$ as real vector spaces (e.g. consider the map $x \mapsto (x_1, \dots, x_{r+s-1})$)

Proposition 2.18. *The image of $L : \mathcal{O}_K^\times \rightarrow H$ is a lattice in H and the kernel $\text{Ker } L = \mu(K)$ is finite.*

Proof. Let $\alpha \in \text{Ker } L$, thus $|\sigma_i \alpha| = 1$ for all i . By Cor. 2.17 α is a root of unity and hence lies in $\mu(K)$, which is finite by Lemma 2.15.

Consider a bounded subset $C \subset H$ containing the origin. C is contained in a box

$$\{(x_1, \dots, x_{r+s}) \in H \mid |x_i| \leq M\}$$

for some bound M . If $L(u) \in C$, then $|\sigma_j u| \leq e^M$ for all j . By Prop. 2.16 there are only finitely many $u \in \mathcal{O}_K^\times$ such that $L(u) \in C$, hence $L(\mathcal{O}_K^\times)$ is discrete and thus it is a lattice. $\square$

We now have

$$\text{Rank } \mathcal{O}_K^\times = \text{Rank}(L(\mathcal{O}_K^\times)) \leq \dim H = r + s - 1,$$

hence we have an upper-bound for the rank of $\mathcal{O}_K^\times$. Now we complete the proof of Dirichlet's theorem by showing the precise rank of $\mathcal{O}_K^\times$ in the following theorem.

Theorem 2.19. *$L(\mathcal{O}_K^\times)$ is a full lattice. Therefore $\text{Rank } \mathcal{O}_K^\times = r + s - 1$.*

Proof. Let $x := (x_1, \dots, x_{r+s}) \in \mathbb{R}^r \times \mathbb{C}^s$ and we define the norm of an element as

$$Nx := x_1 \cdots x_r \cdot x_{r+1} \cdot \bar{x}_{r+1} \cdots$$

Now we have

$$N_{K/\mathbb{Q}}(\alpha) = N(\sigma\alpha)$$

where σ is the Minkowski embedding. Furthermore we have

$$|Nx| = |x_1| \cdots |x_r| \cdot |x_{r+1}|^2 \cdots$$

Let us now consider $x \in \mathbb{R}^r \times \mathbb{C}^s$ such that $|Nx| = 1$ and the set

$$x \cdot \sigma(\mathcal{O}_K) = \{x \cdot \sigma\alpha \mid \alpha \in \mathcal{O}_K\},$$

where " $\cdot$ " denotes the euclidian inner product. By Prop. 2.10 $\sigma(\mathcal{O}_K)$ is a full lattice with covolume

$$\text{Vol } D = 2^{-s} |\Delta|^{1/2}$$

and $x \cdot \sigma(\mathcal{O}_K)$ is also a full lattice with the same covolume.

Consider now $T \subset \mathbb{R}^r \times \mathbb{C}^s$ as in Thm. 2.8. Then there exists a non-zero $\gamma \in \mathcal{O}_K$ such that $x \cdot \sigma\gamma \in T$. Since T is compact, then all its points have bounded coordinates and thus have bounded norm. Consider $x\sigma(\gamma) \in T$, then $N|x\sigma\gamma| \leq M$ for some M that only depends on the choice of T , hence $N_{K/\mathbb{Q}}|\gamma| = N|\sigma\gamma| \leq 2M$. Therefore the principal ideal $\gamma\mathcal{O}_K$ has bounded norm. There are however only finitely many principal ideals satisfying this bound, let us denote them as $(\gamma_1), \dots, (\gamma_t)$. If $x\sigma\gamma \in T$, then $(\gamma) = (\gamma_i)$ for some i , hence there exists a $\varepsilon \in \mathcal{O}_K^\times$ such that $\gamma_i \varepsilon = \gamma$. Then $x\sigma\varepsilon \in \sigma(\gamma_i^{-1})T$. Let

$$T' := \sigma(\gamma_1^{-1})T \cup \cdots \cup \sigma(\gamma_t^{-1})T,$$

which is compact and bounded. Therefore for all x with $|Nx| = 1$ there exists $\varepsilon \in \mathcal{O}_K^\times$ such that $|x_i \sigma_i \varepsilon| \leq M'$ for some M' that is independent of x .

For $r + s - 1 = 0$ the theorem holds. Let $r + s - 1 \geq 1$. For each $1 \leq i \leq r + s$ choose x such that $|Nx| = 1$ and $|x_j| > M'$ for all $j \neq i$. There exists a unit $\varepsilon_i \in \mathcal{O}_K^\times$ such that $|\sigma_j \varepsilon_i| < 1$ for all $j \neq i$ i.e. $\log |\sigma_j \varepsilon_i| < 0$.

Now we shall prove that

$$L(\varepsilon), \dots, L(\varepsilon_{r+s-1})$$

are linearly independent vectors in $L(\mathcal{O}_K^\times)$ i.e. the matrix with rows of the form

$$(\ell_1 \varepsilon_i, \dots, \ell_r \varepsilon_i, 2\ell_{r+1} \varepsilon_i, \dots, 2\ell_{r+s-1} \varepsilon_i)$$

is invertible, where

$$\ell_j \varepsilon_i := \log |\sigma_j \varepsilon_i|.$$

We know $\ell_j \varepsilon_i < 0$ for all $j \neq i$ and

$$\ell_1 \varepsilon_i + \dots + 2\ell_{r+s} \varepsilon_i = 0.$$

Then the sum of i -th row is $-2\ell_{r+s} \varepsilon_i > 0$. Proving the following lemma from linear algebra concludes the proof.

Lemma. *Let (a_{ij}) be a real matrix such that*

(a) $a_{ij} < 0$ for all $j \neq i$

(b) $\sum_j a_{ij} > 0$ for all i

Then it is invertible.

□

The proof of Theorem 2.13 is thus also concluded. Let us now provide some remarks about the system of fundamental units.

How to find the fundamental units

(a) Choose $m \in \mathbb{N}$ and find many $\alpha_i \in \mathcal{O}_K$ such that $N(\alpha_i) = \pm m$. Since there are only finitely many such α_i 's, we will have collisions of the form $(\alpha_i) = (\alpha_j)$, but then $\alpha_i \alpha_j^{-1} \in \mathcal{O}_K^\times$.

(b) Repeat (a) until you have found enough such units, they then generate a subgroup of finite index. But how do we know when we have enough units? We will introduce the notion of regulator, which will help us determine that

Regulator

Let $t := r + s - 1$ and $\varepsilon_1, \dots, \varepsilon_t$ be independent units.

Definition. The *regulator* denoted by $\text{Reg}(\varepsilon_1, \dots, \varepsilon_t)$ is the determinant of a matrix with i -th row of the form

$$\ell \varepsilon_i = (\log |\sigma_1 \varepsilon_i|, \dots, 2 \log |\sigma_{r+1} \varepsilon_i|, \dots).$$

Moreover we define

$$\text{Reg } K := \text{Reg } \mathcal{O}_K^\times = |\text{Reg}(\varepsilon_1, \dots, \varepsilon_t)|,$$

where $(\varepsilon_1, \dots, \varepsilon_t)$ is the system of fundamental units.

Class number formula

Now how do we find $\text{Reg } \mathcal{O}_K^\times$? We will use the class number formula

$$h_K = \frac{w_K |\Delta_K|^{1/2}}{2^r (2\pi)^s \text{Reg } \mathcal{O}_K^\times} \cdot \text{res}_{s=1} \zeta_K(s),$$

where $w_K = \#\mu(K)$ and ζ_K is the Dedekind zeta function. Note that $\mu(K)$ is finite by Lemma 2.15 and $w_K \geq 2$ because we always have $\pm 1 \in \mu(K)$.

The Dedekind zeta function is defined as

$$\zeta_K(s) = \sum_{I < \mathcal{O}_K} N(I)^{-s}$$

and it is an extension of the Riemann zeta function in the sense that choosing $K = \mathbb{Q}$ gives us $\mathcal{O}_K = \mathbb{Z}$ and since all ideals $I < \mathbb{Z}$ are principal of the form (n) for $n \in \mathbb{Z}$ with $N(n) = n$, then we have

$$\zeta_{\mathbb{Q}}(s) = \sum_{n \in \mathbb{Z}} \frac{1}{n^s},$$

which is the Riemann zeta function.

Now since we know how to find h_K from the previous chapter, the only thing that remains is $\text{Reg } \mathcal{O}_K^\times$, which we can obtain from the class number formula.

S-units

Definition. Let K be a number field and S be a finite set of prime ideals in $\mathcal{O}_K$. *Algebraic S-integers* is a set

$$\mathcal{O}_K(S) := \{\alpha \in K \mid \text{ord}_P \alpha \geq 0 \ \forall P \notin S\}.$$

Furthermore we define the set of *algebraic S-units*

$$\mathcal{O}_K^\times(S) := \{\alpha \in K \mid \text{ord}_P \alpha = 0 \ \forall P \notin S\}$$

Exercise. Show, that $\mathcal{O}_K(\emptyset) = \mathcal{O}_K$.

Example. Let $K = \mathbb{Q}$ and $S = \{(2), (3), (5)\}$. Then

$$\mathcal{O}_K^\times(S) = \{\pm 2^k 3^\ell 5^m \mid k, \ell, m \in \mathbb{Z}\}.$$

Theorem. $\mathcal{O}_K^\times$ is finitely generated group of rank $r + s - 1 + \#S$ for all S and a number field K .

3. Cyclotomic fields & FLT

3.1 Cyclotomic fields

From now on let ζ be a primitive n -th root of 1 and consider the number field $K := \mathbb{Q}(\zeta)$. In this section we will aim to understand the ring of integers $\mathcal{O}_K$ of such number fields. Some proofs will not be provided and together with more details can be found in the text (in Czech only) linked bellow.
<https://www.karlin.mff.cuni.cz/~kala/files/TC25.pdf>

Exercise. Show that if ζ is a primitive n -th root of 1, then ζ^a is a primitive n -th root of 1 iff $(a, n) = 1$.
 Now $K = \mathbb{Q}(\zeta)$ is a splitting field for $m_{\zeta, \mathbb{Q}}$ and also it is the splitting field for $x^n - 1 = \prod_{a=1}^n (x - \zeta^a)$.
 Moreover $K/\mathbb{Q}$ is a Galois extension.

Definition. For $n \in \mathbb{N}$ we define the n -th *cyclotomic polynomial*

$$\Phi_n(x) = \prod_{a=1, (a,n)=1}^n (x - \zeta^a) = \prod_{\zeta'} (x - \zeta'),$$

where ζ' is a primitive n -th root of 1.

Remark. (a) $\deg \Phi_n(x) = \phi(n)$, where ϕ is Euler's totient function.

(b) $x^n - 1 = \prod_{d|n} \Phi_d(x)$.

From now on let $G := \text{Gal}(K/\mathbb{Q})$. Elements of G permute the roots of $x^n - 1$. Let $\sigma \in G$, then $\sigma\zeta = \zeta^a$ for some a . Since σ is invertible, then a is also an invertible element mod n , hence $(a, n) = 1$ and thus elements of G map primitive n -th roots to primitive n -th roots. The coefficients of Φ_n are therefore also preserved by elements of G and thus

$$\Phi_n \in \mathbb{Q}[x] \cap \mathcal{O}_{\mathbb{Q}}[x] \implies \Phi_n \in \mathbb{Z}[x].$$

For $(a, n) = 1$ we define a map σ_a such that $\sigma_a\zeta = \zeta^a$. For each $\sigma \in G$ there exists a such that $\sigma = \sigma_a$.

However σ_a is not necessarily well-defined. Consider $f, g \in \mathbb{Q}[x]$, then

$$\sigma_a(f(\zeta)) \stackrel{\text{def}}{=} f(\zeta^a).$$

If we suppose that $f(\zeta) = g(\zeta)$, then for σ_a to be well-defined it is necessary that $f(\zeta^a) = g(\zeta^a)$. Let us examine a simpler case. WLOG let $f(\zeta) = 0$, then we need that ζ^a is also a root of f . Since $f(\zeta) = 0 \iff m_{\zeta, \mathbb{Q}} \mid f$, then it would suffice that $m_{\zeta, \mathbb{Q}}(\zeta^a) = 0$. This is equivalent to Φ_n being the minimal polynomial, hence irreducible.

Consider now the following injective group homomorphism

$$\begin{aligned} \psi : G &\rightarrow (\mathbb{Z}/n)^\times \\ \sigma_a &\mapsto a \end{aligned}$$

This map is surjective (and thus an isomorphism) if and only if Φ_n is irreducible.

We can summarize our findings so far into the following proposition.

Proposition. *TFAE*

- (a) $\psi : G \rightarrow (\mathbb{Z}/n)^\times$ is an isomorphism.
- (b) $[K : \mathbb{Q}] = \phi(n)$
- (c) G acts transitively on primitive n -th roots of 1.
- (d) Φ_n is irreducible in $\mathbb{Z}[x]$.

Proof. Left as an exercise. Try proving equivalences involving (a). □

Theorem 3.1. *Let ζ be a primitive n -th root of 1 and $K = \mathbb{Q}(\zeta)$. Then*

- (a) $[K : \mathbb{Q}] = \phi(n)$
- (b) $\mathcal{O}_K = \mathbb{Z}[\zeta]$ i.e. $1, \zeta, \dots, \zeta^{\phi(n)-1}$ is integral basis.
- (c) if a prime number p ramifies in K , then $p \mid n$. More precisely if $n = p^r m$ with $p \nmid m$ then

$$p\mathcal{O}_K = (\underbrace{P_1, \dots, P_s}_{\text{pairwise distinct}})^{\varphi(p^r)}.$$

Proof. For proof see Milne's Algebraic Number Theory available at:
<http://www.jmilne.org/math/CourseNotes/ant.html> □

Proposition 3.2. *Let $n = p^r$ for p a prime number and let ζ be a primitive n -th root of unity. If we denote $K = \mathbb{Q}(\zeta)$, then*

- (a) $[K : \mathbb{Q}] = \phi(p^r) = p^{r-1}(p-1)$.
- (b) $\mathcal{O}_K = \mathbb{Z}[\zeta]$
- (c) $\pi = 1 - \zeta$ is a prime element in $\mathcal{O}_K$ i.e. $(p) = (\pi)^e$, where $e = \phi(p^r)$. By the EFG theorem (1.18) this means

$$\mathcal{O}_K/(\pi) \simeq \mathbb{Z}/(p).$$

- (d) $\Delta_K = \pm p^c$, where $c = p^{r-1}(pr - r - 1)$.

Proof. Since ζ is a root of $x^n - 1$, then $\zeta \in \mathcal{O}_K$ and $\mathbb{Z}[\zeta] \subset \mathcal{O}_K$. It remains to prove the other direction.

Observation. Let ζ' be some other n -th root of unity, then $\mathbb{Z}[\zeta] = \mathbb{Z}[\zeta']$ and $\mathbb{Q}(\zeta) = \mathbb{Q}(\zeta') = K$.

We have $\zeta^s = \zeta'$ for some $(s, n) = 1$.

$$\begin{aligned} \frac{1 - \zeta'}{1 - \zeta} &= 1 + \zeta + \zeta^2 + \dots + \zeta^{s-1} \in \mathbb{Z}[\zeta] \\ \frac{1 - \zeta}{1 - \zeta'} &\in \mathbb{Z}[\zeta'] = \mathbb{Z}[\zeta], \end{aligned}$$

therefore

$$\frac{1 - \zeta'}{1 - \zeta} \in \mathbb{Z}[\zeta]^\times \subset \mathcal{O}_K^\times.$$

Let us now prove part (a). We have

$$\deg m_\zeta = [K : \mathbb{Q}] \leq \deg \Phi_n = \phi(n).$$

Recall that we can write

$$x^n - 1 = \prod_{d \mid n} \Phi_d(x),$$

hence

$$x^{p^r} - 1 = \Phi_{p^r}(x) \cdot (x^{p^{r-1}} - 1) \tag{3.1}$$

If we denote $t := x^{p^{r-1}}$, we can rewrite

$$\begin{aligned}\Phi_n(x) &= \Phi_{p^r}(x) = 1 + t + \cdots + t^{p-1} \\ \Phi_{p^r}(1) &= p = \prod_{\zeta'} (1 - \zeta') = \prod \frac{1 - \zeta'}{1 - \zeta} (1 - \zeta).\end{aligned}$$

Since $\frac{1-\zeta'}{1-\zeta}$ lies in $\mathcal{O}_K^\times$, then we obtain

$$p = \Phi_{p^r}(1) = u \cdot (1 - \zeta)^e,$$

for some $u \in \mathcal{O}_K^\times$. Denoting $\pi := (1 - \zeta)$ then yields us the equality $(\pi)^e = (p)$. By the EFG theorem (1.18) we then have

$$e \leq [K : \mathbb{Q}] \leq \phi(n) = e,$$

hence the proof of part (a) is concluded. Also since (π) is a prime ideal in $\mathcal{O}_K$, then π is a prime element (exercise), which concludes part (c).

Thanks to the EFG theorem we now also have the following isomorphism

$$\mathcal{O}_K/(\pi) \simeq \mathbb{Z}/(p).$$

Consider the map

$$\begin{aligned}\mathbb{Z}/(p) &\rightarrow \mathcal{O}_K/(\pi) \\ a + (p) &\mapsto a + (\pi)\end{aligned}$$

which is an injection between fields. Since the inertia degree is equal to 1, this map is also an isomorphism, which yields

$$\mathcal{O}_K = \mathbb{Z} + \pi \mathcal{O}_K \tag{3.2}$$

If we now show that

$$\Delta(\mathbb{Z}[\zeta]/\mathbb{Z}) = \Delta(1, \zeta, \dots, \zeta^{n-1}) = \Delta(\mathcal{O}_K/\mathbb{Z}) \cdot (\mathcal{O}_K : \mathbb{Z}[\zeta])^2 = p^k$$

for some k , then we will have $\Delta_K = p^k$ and $(\mathcal{O}_K : \mathbb{Z}[\zeta]) = p^M$ i.e. $p^M \mathcal{O}_K \subset \mathbb{Z}[\zeta]$.

We have

$$\Delta(1, \zeta, \dots, \zeta^{n-1}) = N_{K/\mathbb{Q}}(\Phi'_n(\zeta)).$$

Using (3.1) we obtain

$$\Phi'_n(\zeta) = \frac{n\zeta^{n-1}}{\zeta^{p^{r-1}} - 1}.$$

Furthermore we have

$$\begin{aligned}N(p^r) &= N(p)^r = p^{re} \\ N(\zeta) &= 1.\end{aligned}$$

To determine $N(\zeta^{p^{r-1}} - 1)$, we use the following lemma.

Lemma. $N(1 - \zeta^{p^s}) = \pm p^{ps}$ for all $0 \leq s < r$.

Proof. If $s = 0$, then $N(1 - \zeta)$ is plus or minus the constant term of $\Phi_n(1 - x)$, which is $\Phi_n(1) = p$. The rest of the proof of this lemma as well as the rest of part (d) is left as an exercise. $\square$

What remains to be proven is part (b), so we aim to prove $\mathcal{O}_K = \mathbb{Z}[\zeta]$. From (3.2) we have

$$\begin{aligned}\mathcal{O}_K &= \mathbb{Z}[\zeta] + \pi \mathcal{O}_K \\ \pi \mathcal{O}_K &= \pi \mathbb{Z}[\zeta] + \pi^2 \mathcal{O}_K \\ \mathcal{O}_K &= \mathbb{Z}[\zeta] + \pi \mathbb{Z}[\zeta] + \pi^2 \mathcal{O}_K \\ \mathcal{O}_K &= \mathbb{Z}[\zeta] + \pi^2 \mathcal{O}_K\end{aligned}$$

and hence

$$\mathcal{O}_K = \mathbb{Z}[\zeta] + \pi^j \mathcal{O}_K$$

for all j . Thus we can rewrite it as

$$\mathcal{O}_K = \mathbb{Z}[\zeta] + p^m \mathcal{O}_K$$

for all m . However for $m = M$ we have $p^M \mathcal{O}_K \subset \mathbb{Z}[\zeta]$ and therefore $\mathcal{O}_K = \mathbb{Z}[\zeta]$, which concludes the proof. $\square$

3.2 Regular primes and units

Definition. Let p be an odd prime number and let h_p denote the class number of $\mathbb{Q}(\zeta_p)$. We say that p is *regular* if $p \nmid h_p$.

Remark. It is not known whether there are infinitely many regular primes. However it is known that there are infinitely many irregular primes.

Theorem. *TFAE:*

- (a) p is a regular prime.
- (b) p does not divide the denominator of B_k for all $k = 2, 4, \dots, p-3$, where B_k is the k -th Bernoulli number. Those are numbers that satisfy the following:

$$\frac{t}{e^t - 1} = \sum_{n=0}^{\infty} B_n \cdot \frac{t}{n!}.$$

Definition. Let ζ be a primitive n -th root of unity. We define

$$\mathbb{Q}(\zeta)^+ := \mathbb{Q}(\zeta + \zeta^{-1}),$$

which is the largest *totally real* subfield in $\mathbb{Q}(\zeta)$ i.e. all conjugates of the generator are real or equivalently all embeddings of $\mathbb{Q}(\zeta)^+$ are real.

Lemma 3.3. *Let $K = \mathbb{Q}(\zeta)$ and $n = p^r$. Then for all $u \in \mathcal{O}_K^\times$ there exists $\mu \in \mu(\mathbb{Q}(\zeta))$ and $v \in \mathcal{O}_{\mathbb{Q}(\zeta)^+}^\times$ such that*

$$u = \mu \cdot v.$$

Proof. Proof is left as an exercise. $\square$

3.3 FLT

Theorem 3.4. *Let p be an odd regular prime number. Then there are no $x, y, z \in \mathbb{Z}$ satisfying*

$$x^p + y^p = z^p \quad \text{and} \quad p \nmid xyz.$$

Proof. Assume towards contradiction that there is a solution and that $(x, y, z) = 1$ (they are coprime). For $p = 3$ we have

$$a^3 \equiv 0, \pm 1 \pmod{9}$$

for all a , which leads to a contradiction. Similarly for $p = 5$. We assume then that $p \geq 7$ and WLOG assume $p \nmid x - y$ (exercise).

Consider a polynomial in variable T

$$T^p + 1 = \prod_{i=0}^{p-1} (T + \zeta^i).$$

Now for $T = \frac{x}{y}$ we obtain

$$x^p + y^p = \prod_{i=0}^{p-1} (x + \zeta^i y) = z^p.$$

Let us now view the terms $(x + \zeta^i y)$ in $\mathbb{Z}[\zeta] = \mathcal{O}_{\mathbb{Q}(\zeta)}$.

Lemma 3.5. $x + \zeta^i y \in \mathbb{Z}[\zeta]$ generate pairwise coprime ideals.

Proof of Lemma 3.5. We will prove this lemma by contradiction by assuming that there exists a prime ideal P such that

$$P \mid (x + \zeta^i y), \quad P \mid (x + \zeta^j y)$$

for some $i \neq j$. Then we have

$$P \mid ((\zeta^i - \zeta^j)y).$$

Observe now that $(\zeta^i - \zeta^j) = (\pi) = (1 - \zeta)$ and hence

$$P \mid (\pi y) \text{ and } P \mid (\pi x).$$

Now since x and y are coprime in $\mathbb{Z}$, then they are coprime in $\mathbb{Z}[\zeta]$, therefore $P \mid (\pi)$. However since (π) is a prime ideal, then $P = (\pi)$. Now we have

$$\begin{aligned} \pi = 1 - \zeta &\implies 1 \equiv \zeta^i && \pmod{\pi} \\ \implies x + y &\equiv x + \zeta^i y \equiv 0 && \pmod{\pi} \\ \implies x + y &\in \mathbb{Z} \cap (\pi) = (p) \\ \implies z^p = x^p + y^p &\equiv x + y \equiv 0 && \pmod{\pi} \end{aligned}$$

which is a contradiction. □

Observation 7. (a) $\forall \alpha \in \mathbb{Z}[\zeta] : \alpha^p \in \mathbb{Z} + p\mathbb{Z}[\zeta]$.

(b) Let $\alpha = a_0 + a_1\zeta + \cdots + a_{p-1}\zeta^{p-1}$ where $a_i \in \mathbb{Z}$ and let at least one $a_i \neq 0$. Let $n \in \mathbb{Z}$ be such that $n \mid \alpha$, then $n \mid a_i$ for all i .

Proof of the Observation is left as an exercise.

Consider the following as an equality of ideals in $\mathbb{Z}[\zeta]$

$$(z)^p = \prod_{i=0}^{p-1} (x + \zeta^i y).$$

Ideals $(x + \zeta^i y)$ are pairwise coprime (Lemma 3.5), so their prime factorizations must be disjoint. Moreover each factor must be a p -th power, hence we denote $(x + \zeta^i y) =: A_i^p$. Since A_i^p is principal, then the order of A_i in the class group divides p , therefore it is either 1 or p . By Lagrange's theorem,

the order of A_i also divides h_p and since p is a regular prime, we are left with the order of A_i being 1, which means that A_i is a principal ideal.

Let $A_i =: (\alpha_i)$ for all i and consider for example α_1 :

$$\alpha := \alpha_1 = x + \zeta y = u\alpha^p$$

for some $u \in \mathcal{O}_{\mathbb{Q}(\zeta)}^\times$. By Lemma 3.3 we have

$$u = \mu \cdot v$$

where $\mu = \zeta^r$ for some r . Therefore

$$x + \zeta y = v\zeta^r \cdot \alpha^p,$$

with $v = \bar{v}$, because $v \in \mathbb{Q}(\zeta)^+$ (by Lemma 3.3). Now by the Observation 7 part (a) we have

$$\alpha^p \equiv b \pmod{p\mathbb{Z}[\zeta]}$$

for some $b \in \mathbb{Z}$. From this we have

$$\begin{aligned} x + \zeta y &\equiv v\zeta^r \cdot b \pmod{p\mathbb{Z}[\zeta]} \\ x + \bar{\zeta} y &\equiv v\zeta^{-r} \cdot b \\ \implies vb &\equiv \zeta^{-r}(x + \zeta y) \\ &\equiv \zeta^r(x + \zeta^{-1}y) \end{aligned}$$

which is a contradiction with the Observation 7 part (b). □